



PDF EXPORT

CAESAR II Users Guide

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Piping Input menu: **Edit > Edit Static Load Cases** 

CAESAR II Tools toolbar: **Edit Static Load Cases** 

Displays the [Static Analysis - Load Case Editor Dialog](#) for setting the load cases, wind loads, and wave loads needed for a static analysis. The software recommends an initial set of [Load Cases](#) to accept or modify.

During static analysis, CAESAR II assesses and processes all piping data. The software creates element stiffness matrices and load vectors, and solves for displacements, forces and moments, reactions, and stresses. Additionally, design specific load cases for analysis, select and design spring hangers, and evaluate environmental loads.

Run [error checking](#) prior to performing static analysis. If the model is error free, the software then displays a center of gravity report and generates the analysis data files. It then opens the Preferred Output Reviewer.

If the software finds errors with the piping data, it does not continue analysis. Make corrections and rerun [error checking](#) until no errors exist.

Load Cases

A load case is a group of piping system loads that occur at the same time and are analyzed together. An example of a load case is an operating analysis composed of thermal, dead weight, and pressure loads. Another example is an as-installed analysis of dead weight loads alone.

A load case can also be composed of the combinations of the results of other load cases. For example, a load case can be the difference in displacements between the operating and installed cases.

The software produces a set of reports for a load case, which list restraint loads, displacements and rotations, internal forces, moments, and stresses. Because of piping code definitions for calculation methods and/or allowable stresses, the software also tags load cases with a [Stress Type](#).

[Piping System Loads](#)

[Basic and Combination Load Cases](#)

[Load Case Stress Types](#)

[Recommended Load Cases for Weight, Temperature, and Pressure](#)

[Alternate Sustained \(SUS\) and Occasional \(OCC\) Load Cases](#)

Piping System Loads

Individual piping system loads are used to create load cases. Load types relate to input items found on the Classic Piping Input Dialog. The loads display in the left pane of the [Static Analysis - Load Case Editor Dialog](#).

Load Type	Description	Input items which activate the load
W	Dead weight, including pipe, fluid, and insulation.	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, Fluid Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, Fluid Density, and Rigid Weight.
WNC	Weight, no fluid contents (Pipe weight with no contents). WNC also includes insulation weight <i>except</i> when you set the load case Stress Type to HYD in the Load Case Editor and you set Include Insulation in Hydrotest to False .	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, and Rigid Weight.
WW	Water weight (Weight including pipe and water as the fluid). WW also includes insulation weight <i>except</i> when you set the load case Stress Type to HYD in the Load Case Editor and you set Include Insulation in Hydrotest to False .	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, Water-filled Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, and Rigid Weight.
T1 through T9	Operating temperature (Thermal Set 1 through Thermal Set 9).	Temperature 1 through Temperature 9
P1 through P9	Operating Pressures (Pressure Set 1 through Pressure Set 9).	Pressure 1 through Pressure 9
HP	Hydrostatic test pressure.	Hydro Pressure
D1 through D9	Displacements (Displacement Set 1 through Displacement Set 9).	Displacements (1st Vector) through Displacements (9th Vector)
F1 through F9	Concentrated forces (Concentrated Force Vector 1 through Concentrated Force Vector 9).	Forces/Moments (1st Vector) through Forces/Moments (9th Vector)
WIN1 through WIN4	Wind loads (Wind Load Vector 1 through Wind Vector Load 4).	Wind selected on the Wind Wave tab of the Auxiliary Data panel Wind Shape Factor defined
WAV1 through WAV4	Wave loads (Wave Load Vector 1 through Wave Vector Load 4).	Wave selected on the Wind Wave tab of the Auxiliary Data panel
U1 through U3	Uniform loads (Uniform Load Vector 1 through Uniform Load Vector 3).	Uniform Loads (1st Vector) through Uniform Loads (3rd Vector)
CS	Cold spring.	Material 18 or 19 selected

H	Hanger initial loads.	Hanger Design Data or Predefined Hanger Data
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Basic and Combination Load Cases

Load cases are comprised of one or more load types. A [Stress Type](#), such as sustained (SUS), expansion (EXP), occasional (OCC), operating (OPE), or fatigue (FAT) is also specified as part of the load case definition.

Basic Load Cases

The simplest load case consists of a single load, such as WNC (SUS) for an as-installed weight analysis.

Load cases can also consist of multiple loads added together, such as:

- W+T1+P1+D1+F1 (OPE)
- W+P1+F1 (SUS)

Loads can be preceded by scale factors. This provides several benefits.

- If one loading is a multiple of the other (such as Safe Shutdown Earthquake being two times Operating Basis Earthquake), type only one loading in the Classic Piping Input Dialog. Use this loading in a scaled or unscaled form in the [Static Analysis - Load Case Editor Dialog](#).
- If a loading can be directionally reversible, such as wind or earthquake, type only one loading in the Classic Piping Input Dialog. Use this loading preceded by a + or a - to switch the direction.
- Load Rating Design Factor (LRDF) methods can be implemented by scaling individual load components by their risk-dependent factors. For example:

$$1.05W + 1.1T1 + 1.1D1 + 1.25WIN1$$

Each load case is given an L_n designation as it is created.

Combination Load Cases

Combination load cases combine the results of basic load cases. Define combination load cases after the basic load cases, because basic load cases must be solved first. Define combinations by using the L_n designations, such as:

- L1-L3 (EXP) is a combination case which combines the displacements, forces, and stresses using a combination method that you select.
- L4+L6+L8 (OCC) is a combination case which combines the displacements, forces, and stresses using a combination method that you select.

L_n load cases can be preceded by scale factors.

Algebraic is the most commonly used [Combination Method](#), such as:

- L1-L2 (EXP) is a combination case which combines the displacements, forces, and stresses.
- L4+L6+L8 (OCC) is a combination case which combines the displacements, forces, and stresses.

+ and - signs are unary operators/sign of multiplier. If no value precedes the load for major load cases or the load case for combination cases, then the multiplier is +1.0 or -1.0. If a value precedes the load or the load

case, then the multiplier is +value or -value.

The fatigue (FAT) [Stress Type](#) specifies the expected number of load cycles.

The expansion (EXP) [Stress Type](#) is a combination case that results from subtracting the sustained case from the operating case. The expansion case represents the change in the piping system because of the effect of temperature in the presence of other loads. The restraint status of the operating and sustained cases can be different if there are nonlinear restraints (such as +Y, -Z, any restraint with a gap, and so on) or boundary conditions (such as friction).

The following family of load cases provides an example of algebraic combinations.

Load Case	Designation	Comments
L1	W+T1+P1+H+0.67CS (OPE)	Hot operating. The 0.67scale factor takes credit only for 2/3 of the cold spring.
L2	W1+P1+H+0.67CS (OPE)	Cold operating with cold spring included.
L3	W1+P1+H (SUS)	Traditional sustained case.
L4	WIN1 (OCC)	Wind case. This will be manipulated later to represent average wind 1X, maximum wind 2X (in the positive and negative directions).
L5	L1-L2 (EXP)	Traditional cold to hot expansion case. Use L for load, rather than DS .
L6	L1-L2 (FAT)	Same case evaluated for fatigue at 10,000 cycles.
L7	L1+L4 (OPE)	Hot operating with average wind (in positive direction).
L8	L1-L4 (OPE)	Hot operating with average wind (in negative direction).
L9	L1+2L4 (OPE)	Hot operating with maximum wind (in positive direction).
L10	L1-2L4 (OPE)	Hot operating with maximum wind (in negative direction).
L11	L2+L4 (OPE)	Cold operating with average wind (in positive direction).
L12	L2-L4 (OPE)	Cold operating with average wind (in negative direction).
L13	L2+2L4 (OPE)	Cold operating with maximum wind (in positive direction).
L14	L2-2L4 (OPE)	Cold operating with maximum wind (in negative direction).
L15	L3+L4 (OCC)	Occasional stress case, sustained plus average wind.
L16	L3+2L4 (OCC)	Occasional stress case, sustained plus maximum wind.
L17	L9+L10+L11+L12 (OPE)	Maximum restraint load case. The combination option should be MAX.

NOTE CAESAR II permits the specification of up to 999 load cases for analysis. Copy the model to a new file to specify additional load cases.

Load Case Stress Types

When you initially open the [Static Analysis - Load Case Editor Dialog](#), CAESAR II recommends operating, sustained, and expansion [stress type](#) load cases based on the piping code and loads defined in the model.

Operating (OPE)

Includes the [loads](#) acting on the pipe during hot operation, including primary loads (weight, pressure, and force) and secondary loads (displacements and thermal expansions). Operating cases are used to find hot displacements for interference checking and to find hot restraint and equipment loads. The software combines weight, pressure case, and hanger loads with each thermal load case when recommending operating load cases. For example, the software combines the first displacement set with the first thermal set, the second displacement set with the second thermal set, and so on. Then, the software combines any cold spring loads.

Example

Load Case No.	Definition	Stress Type
L1	W+D1+T1+P1+H	OPE

Sustained (SUS)

Includes the primary force-driven [loads](#) (weight and pressure) acting on the pipe, usually coinciding with the cold as-installed load case. Sustained load cases are used to satisfy the code sustained stress requirements, as well as to calculate as-installed restraint and equipment loads. Sustained load cases are generally built by combining weight with each of the pressure and force sets, and then with any hanger loads.

Example

Load Case No.	Definition	Stress Type
L2	W+P1+H	OPE

Expansion (EXP)

Determines the range between the displacement extremes, usually between the operating and sustained cases. Expansion load cases are [combination](#) cases used to meet expansion stress requirements.

Example 1

Load Case No.	Definition	Stress Type	Combination Method
L3	L1-L2	EXP	Algebraic

Example 2

A model has two operating temperature: one above ambient temperature (L1) and one below ambient temperature (L2). Each operating case has an operating/sustained expansion case. A third expansion cases should be added for the difference between the two operating temperatures.

Load Case No.	Definition	Stress Type
L1	W+D1+T1+P1+H	OPE
L2	W+D2+T2+P1+H	OPE
L3	W+P1+H	SUS

L4	L1-L3	EXP
L5	L2-L3	EXP
L6	L2-L1	EXP

Occasional (OCC)

The software does not recommend occasional load cases. Definition of occasional load cases is your responsibility.

Recommended Load Cases for Weight, Temperature, and Pressure

The [Static Analysis - Load Case Editor Dialog](#) recommends basic and combination load cases based on the the selected piping code and data defined in the model. Edit or delete the recommended load cases, and add new load cases as needed.

 **NOTE** [Recommend Load Cases](#)  restores the default set of load cases.

B31.3, B31.3 Chapter IX

Load Case No.	Definition	Stress Type	Alternate SUS/OCC	Combination Method
L1	W+T1+P1	OPE		
L2	W+P1	SUS	L1	
L3	W+P1	SUS		
L4	L1-L3	EXP		Algebraic

B31.1, B31.4, B31.4 Chapter XI, B31.5, B31.8, B31.9, B31.12 IP, B31.12 PL, ASME NC, ASME ND, NAVY 505, Canadian Z662, BS 806, Swedish 1, Swedish 2, Stoomwezen, RCC-M C, RCC-M D, CODETI, Norwegian TBK-6, FDBR, IGE/TD/12, EN 13480, GPTC/Z380, PD 8010-1, PD 8010-2, ISO 14692, HPFSL, JPI

Load Case No.	Definition	Stress Type	Combination Method
L1	W+T1+P1	OPE	
L2	W+P1	SUS	
L3	L1-L2	EXP	Algebraic

B31.4 Ch IX, B31.8 Ch VIII, Canadian Z662 Chapter 11, DNV *

Load Case No.	Definition	Stress Type
L1	W+T1+P1	OPE
L2	W+P1	SUS

* No expansion stress is calculated for these piping codes.

BS7159, UKOOA *

Load Case No.	Definition	Stress Type
L1	W+T1+P1	OPE

* No expansion or sustained stress is calculated for these piping codes.

NOTES

- Some piping codes perform a code stress check on the operating case and some do not. See US Code Stresses and International Code Stresses in the *CAESAR II Quick Reference Guide* for the equations used by the piping codes to obtain code stress and allowable stress.
- The software creates additional load cases for each Operating Conditions set.
- Delete or modify the recommended load cases if they do not satisfy the analysis requirements.
- If you need assistance with other load case definitions, contact Customer Support.

Alternate Sustained (SUS) and Occasional (OCC) Load Cases

CAESAR II includes a designation for static load cases called an *alternate* load case. The B31.3 2014 code edition merged the requirements of the previous Appendix P into the main body of the standard. This appendix related to stresses due to sustained loads. Because of this change, CAESAR II addresses these additional sustained conditions using the alternate support condition.

Associate a related alternate Sustained (SUS) or alternate occasional (OCC) load case that uses the restraint status from the previous Operating (OPE) load case by selecting [Alternate SUS/OCC](#) in the [Static Analysis - Load Case Editor Dialog](#). The software uses the stiffness values from the OPE case to create the alternate SUS or OCC load case. In addition, the software disables all load case options that are not applicable on the alternate SUS or OCC load case.

B31.3 does not reference an alternate SUS or an alternate OCC load case. However, CAESAR II provides both alternate SUS and alternate OCC because SUS and OCC stresses are both force-based and you may find a need for a similar approach for OCC.

Use this functionality in systems where supports are active in some conditions and inactive in others (such as pipes lifting off supports). In these situations, the changing distribution of sustained loads may influence the difference in strains. The software bases the displacement strain range on the algebraic difference between the calculated positions of the pipe that define the range. In addition to the displacement strain, each calculated position includes the sustained loads present in the condition under evaluation.

The load case generates primary stress corresponding to the restraint configuration for the previous operating (OPE) load case. However, the alternate sustained (SUS) or alternate occasional (OCC) load case is not a valid structural representation of the system. When using the alternate SUS or alternate OCC cases, keep the following in mind:

- These alternate cases determine the stress and the consideration of this stress state in determining the expansion stress range.
- Do not use the restraint loads from an alternate SUS or alternate OCC load case as it is not a true structural representation of the system. CAESAR II does not generate restraint reports for alternate SUS and alternate OCC load cases.

- Do not use an alternate SUS or alternate OCC load case when performing dynamic analysis. Instead, use the corresponding operating case to get the support configuration.
- The default Load Case Template, LOAD.TPL, and an alternate template, LOAD_ALT.TPL, include support for alternate SUS and alternate OCC load cases. Use one of these templates when recommending load cases.
- Alternate SUS provides a simple method to meet the requirements of equation (1b) for the liberal allowable expansion stress range. If you use equation (1a), sustained stress does not affect the allowable expansion stress range.

Example

While alternate SUS properly sets the expansion stress allowable limit in equation (1b), alternate OCC only affects the SUS+OCC load case. You may want to use the following load case stress types:

- L1: OPE (Operating case)
- L2: Alternate SUS (Using the L1 operating support configuration)
- L3: SUS (Standard sustained case with supports set by sustained loads alone)
- L4: OPE (Operating case, or consider including OCC occasional loads)
- L5: Alternate OCC (Using the L4 operating support configuration)
- L6: EXP (Expansion case, defined as L1-L3)
- L7: SUS (Maximum of L2 and L3, to sum with OCC using the Max combination method)
- L8: OCC (L7+L5, using the Scalar combination method)

See [Stress Type](#) and [Combination Method](#).

For more information on loading conditions and support scenarios that result in the greatest sustained load (SL) for each operating condition, see *Appendix S, Example S302*, in the B31.3 code standard.

Recommended Load Cases for Hanger Selection

When the software designs spring hangers, a hanger initial load (H) is provided, and two load cases are required:

- Deadweight carried by the hanger, which is the hanger hot load
- Range of vertical travel

The first load case, traditionally called *restrained weight*, consists of only deadweight (W). For this analysis, CAESAR II includes a rigid restraint in the vertical direction at every location where a hanger is to be sized. The load on the restraint from this analysis is the deadweight that must be carried by the support in the hot condition.

For the second load case, the hanger is replaced with an upward force equal to the calculated hot load, and an operating load case is run. This load case, traditionally called *free thermal*, includes the deadweight and thermal effects, the first pressure set if defined, and any displacements, W+D1+T1+P1. The vertical displacements of the hanger locations, along with the previously calculated deadweights, are then passed

on to the hanger selection routine. After the hangers are sized, the added forces are removed and replaced with the selected supports along with their pre-loading cold loads designated by load component, H. Load component H can be added to the load cases for hanger design for predefined springs. In this case, it would represent the pre-defined operating loads.

CAESAR II then continues with the load case recommendations. A typical set of recommended load cases for a single operating load case spring hanger design is as follows:

Load Case No.	Definition	Stress Type	Comments
L1	W		Weight for hanger loads
L2	W+D1+T1+P1	OPE	Operating for hanger travel
L3	W+D1+T1+P1+H	OPE	Operating, hangers included
L4	W+P1+H	SUS	Sustained load case
L5	L3-L4	EXP	Expansion load case

Cases 1 and 2 generally supply no information to the output reports other than the data found in the hanger tables. Cases 3, 4, and 5 match the recommended load cases for a standard analysis with one thermal and one pressure defined. The displacement combination numbers in case 5 change to reflect the new order. If multiple temperatures and pressures exist in the input, they too display in this set after the second spring hanger design load case.

To calculate the actual cold loads for selected springs, add load case WNC+H before case 3.

If the hanger design criteria of the piping system are set so that the proposed springs must accommodate more than one operating condition, other load cases must be added before case 3. Perform an extra hanger design operating load case for each additional operating load case used to design springs. See [Load Cases with Hanger Design](#).

Load Cases with Hanger Design

When CAESAR II designs spring hangers, two additional load cases are required. The letter H designates the hanger installation load (pre-load) that is always present in a spring hanger.

Load Case No.	Definition	Stress Type	Comments
L1	W	HGR	* HS = Rigid
L2	W+T1+P1	HGR	* HS = Ignore
L3	W+T1+P1+H	OPE	* HS = As Designed
L4	W+P1+H	SUS	* HS = As Designed
L5	L3-L4	EXP **	

*HS is the hanger stiffness specified in the [Static Analysis - Load Case Editor](#).

** Use the algebraic combination method in the [Static Analysis - Load Case Editor](#).

When the model only contains predefined spring hangers, the first two load cases are not needed. However, the letter H is still required in the operating and sustained load cases. Other hanger load cases are required for multiple load case design. In such instances, let CAESAR II recommend the load cases, then add or edit the non-hanger design load cases as necessary.

Static Seismic Load Cases

Because a seismic event is likely to occur while the piping system is in operation, an operating case should have all operating loads plus the seismic load. This load case is then used with the standard operating case to segregate the effect of the seismic load. The seismic load is then combined with the static sustained load case for code compliance considerations.

Load Case No.	Definition	Stress Type
L1	W+T1+P1	OPE
L2	W+T1+P1+U1	OPE
L3	W+T1+P1-U1	OPE
L4	W+T1+P1+U2	OPE
L5	W+T1+P1-U2	OPE
L6	W+T1+P1+U3	OPE
L7	W+T1+P1-U3	OPE
L8	W+P1	SUS
L9	L1-L8	EXP
L10	L2-L1	OCC
L11	L3-L1	OCC
L12	L4-L1	OCC
L13	L5-L1	OCC
L14	L6-L1	OCC
L15	L7-L1	OCC
L16	L8+L10	OCC
L17	L8+L11	OCC
L18	L8+L12	OCC
L19	L8+L13	OCC
L20	L8+L14	OCC
L21	L8+L15	OCC

Load cases 2 through 7 include all the operating loads. The subtracted uniform load vectors reverse the direction of the uniform load applied.

Load cases 10 through 15 signify the segregated occasional load cases. You do not need a code stress check here because these are only part of the final solution for code compliance. Because of this, select the **Suppress** option for the **Output Status**. Also, these combination load cases all use the **Algebraic Combination Method** on the **Load Cases** tab.

Load cases 16 through 21 are all used for code compliance. Add the segregated occasional results to the sustained case results and use either the **Scalar** or **Abs Combination Method**. Both scalar and absolute will give the same code stress results although the displacements, forces, and moments could be different. Because you do not use any results except the stresses for combination cases, the combination method does not matter.

To combine the results of vertical g-loads with horizontal g-loads, a factor is often applied to the vertical g-load component of the combined load. Accomplish this when you type the **Uniform Loads** data on the Classic Piping Input Dialog for the vertical component, or do this directly in the [Static Analysis - Load Case Editor Dialog](#) as shown below. Using the previous example, combine .67 vertical g-load with each horizontal component.

Load Case No.	Definition	Stress Type
L1	W+T1+P1	OPE
L2	W+T1+P1+U1+0.67U2	OPE
L3	W+T1+P1-U1+0.67U2	OPE
L4	W+T1+P1+U1-0.67U2	OPE
L5	W+T1+P1-U1-0.67U2	OPE
L6	W+T1+P1+U3+0.67U2	OPE
L7	W+T1+P1-U3+0.67U2	OPE
L8	W+T1+P1+U3-0.67U2	OPE
L9	W+T1+P1-U3-0.67U2	OPE
L10	W+P1	SUS
L11	L1-L10	EXP
L12	L2-L1	OCC
L13	L3-L1	OCC
L14	L4-L1	OCC
L15	L5-L1	OCC
L16	L6-L1	OCC
L17	L7-L1	OCC
L18	L8-L1	OCC
L19	L9-L1	OCC
L20	L10+L12	OCC
L21	L10+L13	OCC
L22	L10+L14	OCC
L23	L10+L15	OCC
L24	L10+L16	OCC
L25	L10+L17	OCC
L26	L10+L18	OCC
L27	L10+L19	OCC

Sometimes you need to combine the horizontal and vertical components of seismic loading. From the [Static Analysis - Load Case Editor Dialog](#), set up the static seismic load cases as shown in the first example, then combine the segregated horizontal and vertical load cases together using the **SRSS** [Combination Method](#). Add these results to the sustained case.

Load Case No.	Definition	Stress Type
L1	W+T1+P1	OPE

L2	W+T1+P1+U1	OPE
L3	W+T1+P1-U1	OPE
L4	W+T1+P1+U2	OPE
L5	W+T1+P1-U2	OPE
L6	W+T1+P1+U3	OPE
L7	W+T1+P1-U3	OPE
L8	W+P1	SUS
L9	L1-L8	EXP
L10	L2-L1	OCC *
L11	L3-L1	OCC *
L12	L4-L1	OCC *
L13	L5-L1	OCC *
L14	L6-L1	OCC *
L15	L7-L1	OCC *
L16	L10+L12	OCC **
L17	L10+L13	OCC **
L18	L11+L12	OCC **
L19	L11+L13	OCC **
L20	L14+L12	OCC **
L21	L14+L13	OCC **
L22	L15+L12	OCC **
L23	L15+L13	OCC **
L24	L8+L16	OCC ***
L25	L8+L17	OCC ***
L26	L8+L18	OCC ***
L27	L8+L19	OCC ***
L28	L8+L20	OCC ***
L29	L8+L21	OCC ***
L30	L8+L22	OCC ***
L31	L8+L23	OCC ***

* Use the **Algebraic** [Combination Method](#).

** Use the **SRSS** [Combination Method](#).

*** Use the **Abs** or **Scalar** [Combination Method](#).

NOTE Change the operating load cases that include seismic loads to OCC for piping codes that do not perform a sustained code stress check. Use these cases for code compliance. The combination cases are not needed in such cases.

Load Cases with Pitch and Roll

There is often platform movement, or relative movement, between two platforms with inter-connected piping, in an offshore piping system. This also applies to FSPO and other shipboard piping systems. Apply

the pitch and roll displacements to CNodes on each affected restraint. Use displacement vectors not already in use to describe thermal displacement boundary conditions, usually with a positive (+) displacement and a negative (-) displacement to describe the peak pitch and roll conditions. Look at the state of the platform at its peaks to determine the worst two conditions for relative displacement between piping separated by the largest distance along the line of wave travel. D3 and D4 describe two peak pitch conditions. D1 is a thermal displacement.

Load Case No.	Definition	Stress Type
L1	W+T1+D1+D3+P1	OPE
L2	W+T1+D1+D4+P1	OPE
L3	W+P1	SUS
L4	L1-L3	EXP *
L5	L2-L3	EXP *

* Use the **Algebraic Combination Method** in the [Static Analysis - Load Case Editor Dialog](#).

A fatigue analysis is likely because of the large number of displacement cycles common in pitch and roll situations. Select the appropriate fatigue curve on the first piping input under the Allowable Stresses **Auxiliary Data** tab on the Classic Piping Input Dialog. Add the following cases to the previous example. Enter the number of cycles for each pitch condition for fatigue stress type (FAT).

Load Case No.	Definition	Stress Type	No. of Cycles
L6	L1-L3	FAT	21,000,000
L7	L2-L3	FAT	21,000,000

21,000,000 represents 21 million load cycles during the life of the piping system. Use the number of cycles that you would expect to occur during the life of such a storm for large displacements, such as those that occur during a 1-year, 30-year, or 100-year event. Multiply this number by the number storms likely to happen during the lifetime of the piping system.

Load Cases with Thermal Displacements

Thermal displacements are generally associated with specific operating conditions. D1 is applied with T1, D2 to T2, and so on. When one temperature is below ambient, and one is above ambient, determine the full expansion stress range.

Load Case No.	Definition	Stress Type	Comments
L1	W+T1+D1+P1	OPE	
L2	W+T2+D2+P1	OPE	
L3	W+P1	SUS	
L4	L1-L3	EXP	* Effects of D1 and T1
L5	L2-L3	EXP	* Effects of D2 and T2
L6	L1-L2	EXP	* Full expansion stress range

* Use the **Algebraic Combination Method** in the [Static Analysis - Load Case Editor Dialog](#).

NOTE Include the thermal displacements in the operating cases as shown for piping codes with no expansion stress computation.

Load Cases with Thermal Displacements and Settlement

Use a CNode for settlement on any affected restraints. This CNode must be a node number that is not used elsewhere in the model. Place the settlement on the CNode using a displacement vector that is not already used for thermal displacements. This example uses D3 to describe restraint settlement. Settlement is evaluated as an expansion load because it is strain related with a half-cycle.

Load Case No.	Definition	Stress Type	Comments
L1	W+T1+D1+D3+P1	OPE	
L2	W+T2+D2+D3+P2	OPE	
L3	W+P1	SUS	
L4	W+P2	SUS	
L5	L1-L3	EXP	* Effects of D1 and T1 and settlement
L6	L2-L4	EXP	* Effects of D2 and T2 and settlement
L7	L1-L2	EXP	* Full expansion stress range between OPE1 and OPE2

* Include the thermal and settlement displacements in the operating cases as shown for piping codes with no expansion stress computation.

Error Check / Start Run

Piping Input menu: **File > Error Check** 

CAESAR II Tools toolbar: **Start Run** 

Shortcut key: **CTRL+E**

Saves the input and starts the error checking procedure. When the error check completes, the [Errors and Warnings Tab](#) displays the results.

Select the options arrow ▼ to control the displayed errors and warnings:

Show All

Displays all errors, warnings, and notes.

Show Fatal Errors Only

Displays fatal errors without displaying warnings and notes.

Limit Repeating Messages

Stops displaying duplicate messages when the quantity exceeds the value of **Set Message Repeat Limit**.

Set Message Repeat Limit

Displays the **Set Display Limit** dialog, where you set the number of duplicate messages to repeat.

Complete the error check of Piping Input before performing [Static Analysis](#). When error checking completes, the software creates the required analysis data files. If the piping input changes, CAESAR II does not allow an analysis to take place until you successfully rerun the error checker.

[Errors and Warnings Tab](#)

[Classic Piping Input Tab](#)

[MDMT Report Tab](#)

 **NOTE** Select [Batch Run](#) to check, analyze, and display results with one command.

Errors and Warnings Tab

Displays the total numbers of errors, warnings, or notes for the current job.

- Resolve all errors for the software to run the analysis.
- Warnings and notes are for reference and do not prevent analysis.

Double-click a row of the message. The software zooms the graphic view to the affected element. Select the tabs at the bottom of the window to alternate between the Classic Piping Input Dialog and the **Errors and Warnings** dialog.

Double-click the column headers to sort the messages by **Message Type**, **Message Number**, or **Element/Node** Number. Select File Menu > Print to print the messages.

Select and Copy one or more rows and then paste into other software, such as Microsoft Excel. Select a row, press SHIFT and select multiple rows, or press CTRL+A to select all rows.

Errors 1 Warnings 2 Notes 2				
	Message Type	Message Number	Element/Node Number	Message Text
1	ERROR	2E	35-40	The length of element 35 TO 40 is ZERO.
2	WARNING	33E	30-35	There is a GEOMETRIC INTERSECTION at node 35 and there was NO user defined INTERSECTION TYPE. Standard pipe flexibilities and SIF's = 1.0 will be used unless overridden by the user on each element framing into the intersection.
3	WARNING	33E	35-40	There is a GEOMETRIC INTERSECTION at node 40 and there was NO user defined INTERSECTION TYPE. Standard pipe flexibilities and SIF's = 1.0 will be used unless overridden by the user on each element framing into the intersection.
4	NOTE	69E	110-115	Number of HANGERS in this job = 2 Number of HANGERS to design = 2 Number of Pre-Designed HANGERS = 0
5	NOTE	70E	110-115	The LOAD CASES below should be defined at analysis level to properly perform the HANGER DESIGN: COMBINATION LOAD CASE 1 WEIGHT LOADS COMBINATION LOAD CASE 2 SPR. DESIGN OPERATING CASE #1 COMBINATION LOAD CASE 3 Any cases with the designed hangers installed.

Error Message

The software reports an error when the analysis cannot continue. Errors can be caused by a problem in the model, such as a piping element with no defined length. Correct all errors before the software allows you to continue the analysis.

Warning Message

The software reports a warning when there is a problem that can be overcome using some assumptions. An example of this is the wall thickness of an element that is insufficient to meet the minimum wall thickness for the given pressure (hoop stress). Carefully review warnings. Correcting warnings is not required before performing an analysis.

Note Message

The software reports a fact related to the model. An example of a note is the number of hangers to be designed by CAESAR II or the Center of Gravity report generated after analysis. Note messages are just informational, and there is no action required.

Classic Piping Input Tab

Displays the Classic Piping Input Dialog.

MDMT Report Tab

Displays information for the minimum design metal temperature (MDMT) check when analyzing a model using the B31.3 or B31.3 Chapter IX piping codes.

OK

The operating temperature is greater than or equal to the MDMT.

Impact

The operating temperature is less than the MDMT, and impact testing is required.

Unknown

The material does not have the MDMT specified or the element's pipe thickness falls outside the ranges specified in Fig. 323.2.2A.

Example Report

CAESAR II B31.3 UNADJUSTED MINIMUM DESIGN METAL TEMPERATURE (MDMT) WITHOUT IMPACT TESTING (TABLE A-1 / FIG. 323.2.2A)							
FROM	TO	PIPE THK (in.)	MAT/CURVE	CODE	MIN[Ta, T1~T9] (F)	MDMT (F)	TEST STATUS
10	20	0.600	110/A	B31.3	40.00	40.60	Impact
20	30	0.600	110/A	B31.3	65.00	40.60	OK
30	40	3.300	110/A	B31.3	65.00	NA	Unknown

Batch Run

Piping Input menu: **File > Batch Run** 

CAESAR II Tools toolbar: **Batch Run** 

Static Analysis - Load Case Editor Dialog: **Analyze Load Cases** 

Shortcut key: **CTRL+R**

[Checks](#) the input data, [analyzes](#) the system, and displays the output reviewer (New Analysis Reviewer or Static Output Processor) in one step. New Analysis Reviewer is the default output reviewer.

NOTES

- Select View Static Results  at any time to display the classic Static Output Processor.
- Change the Preferred Output Reviewer in the Configuration Editor.

★ **IMPORTANT** The software assumes that the load cases associated with the current job do not need to change, and that the default account number (if Accounting is active) is correct. These criteria are usually met after the first pass through the analysis.

Static Analysis - Load Case Editor Dialog

Main window ribbon: **Home > Analysis > Static Analysis** 

Main window ribbon: **Analysis > Piping > Static Analysis** 

In Piping Input:

Piping Input menu: **Edit > Edit Static Load Cases** 

CAESAR II Tools toolbar: **Edit Static Load Cases** 

After successfully running [Error Check](#)  in Piping Input, defines the [Load Cases](#) for static analysis.

The **Static Analysis - Load Case Editor** dialog lists many details about the input, including the following:

- Available loads that are defined in the input.
- Available stress types.
- Current load cases offered for analysis.

CAESAR II provides [Recommend Load Cases](#) if the job is entering static analysis for the first time. The list displays loads saved during the last session if the job has been run previously.

Options (on all tabs)



Save

Saves the load case and environmental data file.



[Analyze load cases](#)

Runs the static analysis.

Tabs

[Load Cases Tab \(Static Analysis - Load Case Editor Dialog\)](#)

[Wind Loads Tab \(Static Analysis - Load Case Editor Dialog\)](#)

[Wave Loads Tab \(Static Analysis - Load Case Editor Dialog\)](#)

Load Cases Tab (Static Analysis - Load Case Editor Dialog)

Controls options for editing load cases. Load cases display in a list view by default. Define up to 999 [load cases](#).

List View Functions

Drag one or more loads in the [Loads Defined in Input](#) pane on the left to a [Definition](#) cell in the list to create an algebraic load case. Load names can also be typed in the [Definition](#) cell.

Drag one or more load cases (**L1**, **L2**, and so on) to a [Definition](#) cell in the list to create an combination load case. Load case names can also be typed in the [Definition](#) cell. The software prompts you to select the [Combination Method](#).

Specify all basic (non-combination) load cases before creating combination load cases. This is true for both user-defined and edited load cases.

After creating a load case, edit the load case properties, such as [Stress Type](#), and [Name](#).

Drag a column header and drop it here to group by that column

Groups the list by one or more columns. Select a column heading and drag it above the column headings.

▼ Filter

Provides options to filter load cases by each column property. Select ▼ in a column heading and select the filter options.

Static columns

Keeps the [Load Cases](#), [Exclude](#), and [Definition](#) columns visible when scrolling the list view to the right. Drag other columns to the left of the static column separator to make them static.

Column re-ordering

Drag the column heading to change the order of the **List** view columns.

Load Case Editor context menu

Right-click any load case to access the following options: **Insert Row Above**, **Insert Row Below**, **Delete**, **Copy**, **Paste Over**, and **Insert Copied Above**.

Group Edit

Displays a group load case. Select multiple load cases and edit common properties. See [Edit multiple load cases](#).

 **NOTE** To return to the load case grid, select **List**.

List

Changes the **Group Edit** view back to the list view.

Options



Save

Saves the load case and environmental data file.



Add a load case

Inserts a blank line following the selected load case line in the list. If you do not select a row, the software adds the load case at the end of the list.

NOTES

- Right-click > **Insert Before** to insert a blank line before the selected load case line in the list.
- Right-click > **Insert After** to insert a blank line following the selected load case line in the list.



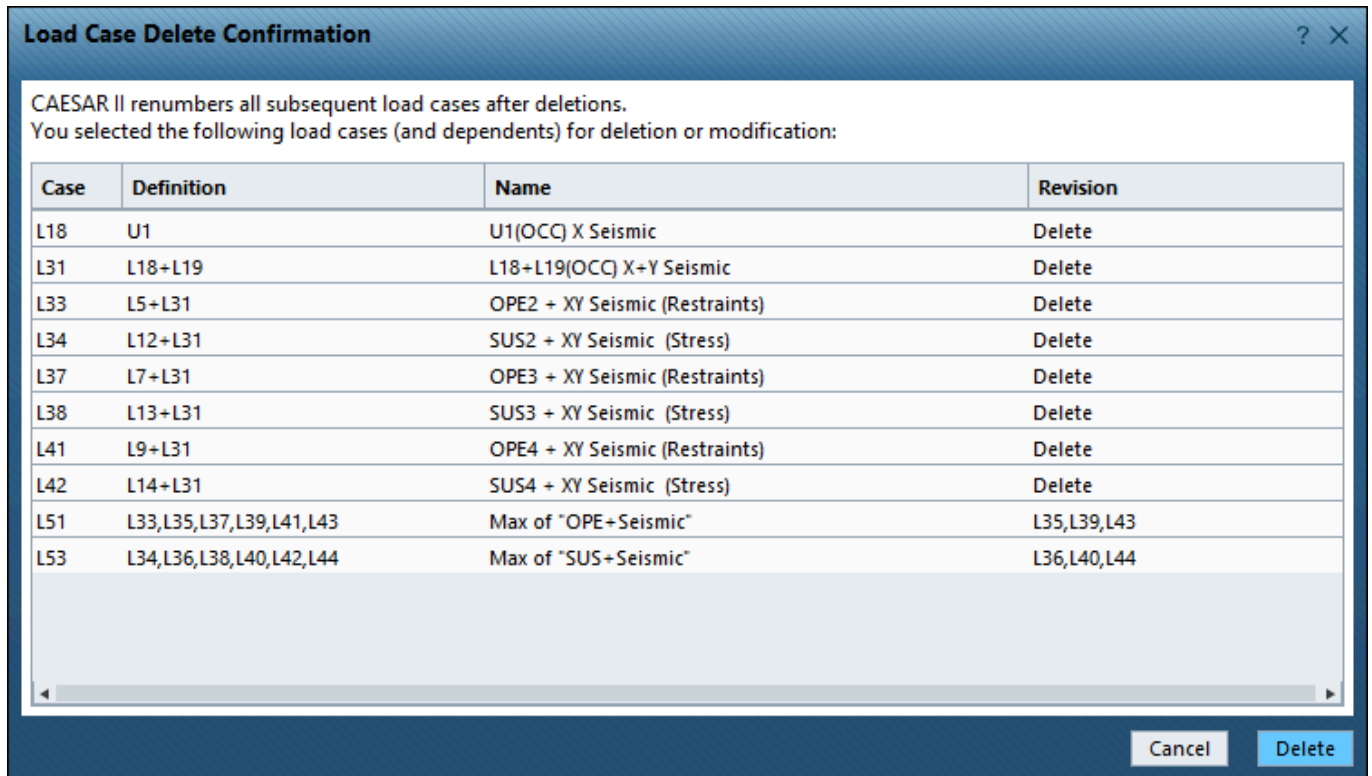
Delete a load case

Deletes one or more selected load cases. SHIFT-click to select multiple adjacent rows. CTRL-click to select multiple non-adjacent rows.

When you delete a load case, any dependent [combination load cases](#) are affected. The software deletes the entire combination load case if it only contains two primitives. However, for combination load cases with three or more primitives, the software modifies the load case to remove only the applicable primitive.

When you delete an operating (OPE) load case, an alternate sustained or occasional load case based on the operating case (SUS or OCC, with [Alternate SUS/OCC](#) selected) is also deleted.

Review and confirm the load cases that will be deleted or revised in the **Load Case Delete Confirmation** window. The **Revision** column displays the action to be performed for each load case. After reviewing the load cases, press **Delete** to complete the process.



NOTE Also right-click > **Delete**.

Copy and Paste Options

Copy and paste one or more rows in the grid to create new load cases. Select the line number in the left-most column of the grid (such as L1) to select a row. SHIFT-click to select multiple continuous rows. CTRL-click to select multiple non-continuous rows.



Copy

Copies one or more selected rows to the clipboard.

NOTE Also right-click > **Copy** or press CTRL-C.



Paste > Paste Copied

Pastes one or more copied rows at the selected row, also overwriting the selected and following rows.

NOTE Also right-click > **Paste > Paste Copied** or press CTRL-V.



Paste > Insert Copied

Pastes one or more copied rows at the selected row, inserting the copied rows above the selected rows. When you select multiple rows, the software inserts the copied rows above the first selected row.

NOTE Also right-click > **Paste > Insert Copied** or press CTRL-I.



[Recommend Load Cases](#)

Replaces the current load cases with the CAESAR II recommended load cases.

[Import Load Cases](#)

Copies the load cases from a .c2 or .j file. The units and load types of the copied file must match those of the current file.

Import Excel File

Copies the load cases from a Microsoft Excel .xlsx file. The units and load types of the copied file must match those of the current file. Select the **Import load cases**  flyout to display this option.

NOTES

- The fields of the .xlsx file must match the **Static Analysis - Load Case Editor** fields.
- The software provides template files, *Load_Cases_Template.xltx* and *Load_Cases_Template.xlsx*, in the C:\ProgramData\Intergraph CAS\CAESAR II\<current version>\System folder.

[Export to Microsoft Excel File](#)

Exports the load cases to a Microsoft Excel .xlsx file. Use the file to import load cases into another model.

 **NOTE** By default, the software saves the .xlsx file in the same folder as the piping model.

[Analyze load cases](#)

Runs the static analysis.

NOTES

- Data can be copied from an Excel .xlsx file.
- You can reorder the columns in the list, but you cannot copy, change the column order, and then try to paste in the new order. If you reorder columns, copy again in the new column order, and then paste.
- Non-continuous rows can be copied. The software pastes the rows as continuous rows.
- If you filter the load cases, you can copy one or more rows, but you cannot paste until you clear the filter.
- When defining a fatigue (FAT) stress type for a load case, enter the number of anticipated **Load Cycles** for that load case.
- When deleting load cases, the software automatically rennumbers all subsequent load cases.

Recommend Load Cases

 Displays the [Recommended Load Cases Dialog](#), which suggests the [load cases](#) and stress types that satisfy the model parameters and the basic requirements of the selected piping code.

Use this command to reset customized load cases to the software recommended set.

Recommended Load Cases Dialog

Displays a list of load cases that CAESAR II recommends satisfying the expansion and sustained code compliance requirements of the selected piping code.

Accept

Accept the recommended load cases and close the dialog. The recommended load cases replace any load cases that were on the [Static Analysis - Load Case Editor Dialog](#).

 **NOTE** The load cases can be modified.

Cancel

Closes the dialog and returns to the [Static Analysis - Load Case Editor Dialog](#) without making changes to the load cases.

Import Load Cases



Imports load cases from a CAESAR II file.

 **NOTE** Make sure the units match between the two jobs files prior to importing. When you import load cases from another job, verify the load case values, wind data, and wave data to ensure data integrity.

Loads Defined in Input

Displays the loads available based on parameters defined in Piping Input. Displays the loads in the left pane of the [Static Analysis - Load Case Editor Dialog](#). For example, if **T2** displays, then operating Temperature 2 is defined in the model. If **T2** does not display then the model does not include a second operating temperature.

Load Type	Description	Input items which activate the load
W	Dead weight, including pipe, fluid, and insulation.	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, Fluid Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, Fluid Density, and Rigid Weight.
WNC	Weight, no fluid contents (Pipe weight with no contents). WNC also includes insulation weight <i>except</i> when you set the load case Stress Type to HYD in the Load Case Editor and you set Include Insulation in Hydrotest to False .	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, and Rigid Weight.

WW	Water weight (Weight including pipe and water as the fluid). WW also includes insulation weight <i>except</i> when you set the load case Stress Type to HYD in the Load Case Editor and you set Include Insulation in Hydrotest to False .	Pipe Weight, Insulation Weight, Refractory Weight, Cladding Weight, Water-filled Weight, and Rigid Weight, calculated from: Diameter, Material, Insul Thk, Insulation Density, Refract Thk, Refract Density, Clad Thk, Cladding Density, and Rigid Weight.
T1 through T9	Operating temperature (Thermal Set 1 through Thermal Set 9).	Temperature 1 through Temperature 9
P1 through P9	Operating Pressures (Pressure Set 1 through Pressure Set 9).	Pressure 1 through Pressure 9
HP	Hydrostatic test pressure.	Hydro Pressure
D1 through D9	Displacements (Displacement Set 1 through Displacement Set 9).	Displacements (1st Vector) through Displacements (9th Vector)
F1 through F9	Concentrated forces (Concentrated Force Vector 1 through Concentrated Force Vector 9).	Forces/Moments (1st Vector) through Forces/Moments (9th Vector)
WIN1 through WIN4	Wind loads (Wind Load Vector 1 through Wind Vector Load 4).	Wind selected on the Wind Wave tab of the Auxiliary Data panel Wind Shape Factor defined
WAV1 through WAV4	Wave loads (Wave Load Vector 1 through Wave Vector Load 4).	Wave selected on the Wind Wave tab of the Auxiliary Data panel
U1 through U3	Uniform loads (Uniform Load Vector 1 through Uniform Load Vector 3).	Uniform Loads (1st Vector) through Uniform Loads (3rd Vector)
CS	Cold spring.	Material 18 or 19 selected
H	Hanger initial loads.	Hanger Design Data or Predefined Hanger Data

Load Cases

Displays the load case numbers in the column between [Loads Defined in Input](#) and [Exclude](#). Load cases are numbered in sequential order as **L1**, **L2**, and so on.

Exclude

Removes a selected load case from static analysis. Select to exclude a load case. By default, the software includes all load cases, with the **Exclude** option cleared.

Definition

Specifies the [Load Cases](#) available for analysis. Build basic load cases by dragging load components from [Loads Defined in Input](#) list on the left and dropping them on the **Definition** box or by typing in the values. Alternatively, type a previously created load case (such as L1) to create a combination load case.

NOTES

- The piping code and the options selected in Piping Input determine what load case values display in this list.
- **Definition** text cannot exceed 132 characters.

Name

Describes the load case. Define names or labels for load case definitions for identification purposes when reviewing the output. Clear the **Name** field to display the CAESAR II load case definition name.

 **NOTE** Name text cannot exceed 132 characters.

User-defined load case names appear in the **Load Case Report**. For more information, see [Control analysis results](#). Use these names in place of the default load case definition name anywhere in the [Static Analysis - Load Case Editor Dialog](#).

Stress Type

Displays the stress type for each load case. Stress type determines the stress calculation method and the allowable stress to use.

OPE

Operating case. For B31.1 and B31.3 (and similar codes) this case is not a code compliance case. The software does not report allowable stresses.

SUS

Sustained case.

EXP

Expansion case.

OCC

Occasional case.

FAT

Fatigue case.

 **NOTE** You must also specify the number of [Load Cycles](#) for load cases with a FAT stress type.

HGR

Spring hanger design case. These are load cases that CAESAR II uses internally to design and select spring hangers. Results are not available for these cases.

HYD

Hydro test case. Select hanger status. For a hydrotest case, the default hanger status is rigid or locked.

CRP

Creep Loading case. Code standards such as EN-13480 define a creep stress range for operating conditions, which is defined by the life of the material. In CAESAR II, CRP is a scalar combination of one SUS case and one EXP case. The software sets [Output Type](#) to **Stress**.

The software calculates CRP stresses according to EN-13480. If you specify additional load multipliers, the software applies these as additional scale factors. Other codes also use the EN-13480 method.

NOTES

- You must manually add a CRP case.
- The default combination method is **Scalar**. You can also set [Combination Method](#) to **MAX**.

K1P

KHK Level 1 (seismic code) primary longitudinal stress for the HPGSL and JPI piping codes. The longitudinal stress is due to pressure, weight, and design seismic force.

NOTE The software treats this stress type as **OCC** for other piping codes.

K1SR

KHK Level 1 (seismic code) secondary cyclic stress range for the HPGSL and JPI piping codes. The cyclic stress range is due to design force and support movement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2P

KHK Level 2 (seismic code) primary longitudinal stress for the HPGSL and JPI piping codes. The longitudinal stress is due to internal pressure, weight, seismic force, and response displacement.

NOTE The software treats this stress type as **OCC** for other piping codes.

K2SA

KHK Level 2 (seismic code) secondary cyclic stress amplitude for the HPGSL and JPI piping codes. The cyclic stress amplitude is due to seismic force and response displacement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2SR

KHK Level 2 (seismic code) secondary cyclic stress range for the HPGSL and JPI piping codes. The cyclic stress range is due to seismic force and response displacement.

NOTE The software treats this stress type as **EXP** for other piping codes.

K2L

KHK Level 2 (seismic code) liquefaction for the HPGSL and JPI piping codes. Liquefaction is the angular displacement corresponding to a maximum equivalent plastic strain of 5% (in degrees).

NOTE The software treats this stress type as **EXP** for other piping codes.

LMST

For DNV 2017, only use the **LMST** limit state for combined load cases where local buckling for the combined loading criteria is calculated. For more information, see section 5.4.6 of DNVGL-ST-F101.

MDMT

For B31.3, the minimum design metal temperature.

Alternate SUS/OCC

Indicates that the load case is an [alternate sustained \(SUS\) or alternate occasional \(OCC\) load case](#). An alternate SUS/OCC load case is associated with the preceding operating (OPE) load case.

Due to changes in the B31.3 code for the 2014 edition, the primary stress in each position of the piping system must be evaluated. An alternate SUS/OCC load case uses the restraint status from the previous OPE load case to evaluate the stresses induced by primary loads.

When you create a load case that follows an OPE load case and select a **Stress Type** of **SUS** or **OCC**, the software displays **Alternate SUS/OCC**. Select to indicate that the case is an alternate SUS or alternate OCC load case.

CAESAR II displays the number of the related OPE load case when you select **Alternate SUS/OCC**.

Load Cycles

Indicates the anticipated number of applications of this load on the system for load cases using the Expansion (**EXP**) or Fatigue (**FAT**) [stress types](#). The value is typically in the thousands or greater.

The software uses the **Load Cycles** value to determine the allowable stress from the fatigue curve for the material or the cyclic stress range reduction factor F1, F2, ... F9 for an **EXP** case. The value calculated from **Load Cycles** overrides the value defined for F1, F2, ... F9.

When a value is not defined, the software assumes the model has no reduction and uses a default value of 7000 for **Load Cycles**, equivalent to a cyclic stress range reduction factor F1, F2, ... F9 of 1.0.

NOTES

- For static cases, the software calculates stresses at full range.
- For dynamic cases, the software calculates stresses at half range, which is the amplitude of the full stress range.

B31.1 only

For B31.1 load cases with an Expansion (**EXP**) stress type, define **Load Cycles** as 1.0 to indicate that the stress range is caused by noncyclic support movements. In this situation, the software replaces the allowable stress range value, S_a , with a special noncyclic allowable calculation according to the B31.1 code. The software continues to calculate the displacement stress range, S_e .

Output Status

Controls the disposition of the load case results.

Keep

Displays the results. Use **Keep** when the load case is producing results that you want to review. The default for all new cases (except for HGR load cases) is **Keep**.

Suppress

Hides the results. Use **Suppress** for artificial cases such as the preliminary hanger cases, or intermediate construction cases.

★ **IMPORTANT** Load cases used for hanger design, that is, the weight load case and hanger travel cases designated with the stress type HGR, must be designated as **Suppress**.

Example

A wind only load case is set to **Suppress** because it is only used in other combination load cases and has no value as a standalone load case.

Output Type

Designates the type of results available for load cases that have a **Keep** status. Use this field to help minimize clutter on the output and to ensure that only meaningful results are retained. The available options are:

Disp/Force/Stress

Provides displacements, restraint loads, global and local forces, and stresses. This is a good choice for operating cases when designing to a code which does a code check on operating stresses, because the load case is of interest for interference checking (displacements) and restraint loads at one operating extreme (forces).

Disp/Force

Provides displacements restraint loads, global and local forces. This is a good choice for OPE cases when designing for those codes which do not do a code check on OPE stresses.

Disp/Stress

Provides displacements and stresses only.

Force/Stress

Provides restraint loads, global and local forces, and stresses. This is a good choice for the Sustained (cold) case, because the load case would be of interest for restraint loads at one operating extreme (forces), and code compliance (stresses).

 **NOTE** FR combination loads cases developed under previous versions of CAESAR II are converted to this force/stress type.

Disp

Provides displacements only.

Force

Provides restraint loads, global, and local forces only.

Stress

Provides stresses only. This is a good choice for a sustained plus occasional load case (with Abs combination method), because this is an artificial construct used for code stress checking purposes.

NOTE ST combination load cases developed under previous versions of CAESAR II are converted to this stress type.

Combination Method

Specifies the combination method to use only for combination cases (containing other load cases, such as L1 and L2).

NOTE Load case results are multiplied by any associated scale factors before performing the combination and comparison.

The available methods are:

Algebraic

Indicates a signed algebraic combination of displacement and force level. This method combines the displacement vectors and the force vectors algebraically and then calculates the stresses from the combined forces. Displacements are the algebraic combination of the displacement vectors. Forces are the algebraic combination of the force vectors. Stresses are not combined. Stresses are calculated from the algebraically combined forces. The **Algebraic** method is typically used to calculate EXP code stresses.

New combination cases automatically default to **Algebraic**.

Algebraic combinations can be built only from basic load cases. Basic load cases are non-combination load cases or other load cases built using the Algebraic combination method.

NOTE The obsolete CAESAR II combination methods DS and FR used an algebraic combination method. Therefore, load cases built in previous versions of CAESAR II using the DS and FR methods are converted to **Algebraic**.

Scalar

Indicates a signed combination of displacement, force, and stress level. This method combines the displacement vectors, force vectors, and stress scalars. Displacements are the algebraic combination of the displacement vectors. Forces are the algebraic combination of the force vectors. Stresses are the scalar combination of the stress scalars.

The combination of displacements and forces are the same for **Algebraic** and **Scalar** methods. The combinations of stress levels are different between **Algebraic** and **Scalar** methods because the stresses are calculated from the combined forces in the **Algebraic** method and summed in the **Scalar** method.

Example

Load Case 1: bending stress = 100 psi, due to X-moment

Load Case 2: bending stress - 100 psi, due to Z-moment

Algebraic (vectorial) sum = square root of $(100*100 + 100*100) = 141.4$ psi

Scalar sum = $100 + 100 = 200$ psi

Scalar is typically used to sum (SUS + OCC) code stresses.

NOTES

- The obsolete CAESAR II combination methods ST used a scalar combination method. Therefore, load cases built in previous versions of CAESAR II using the ST method are converted to **Scalar**.

For DNV, only **Abs** and **Scalar** are allowed.

SRSS

Indicates a combination of the square root of the sum of the squares of quantities, such as the displacements of the forces or the stresses. Displacements are the square root of the sum of the squares of the displacements of all cases included in the combination. Forces are the square root of the sum of the squares of the forces of all cases included in the combination. Stresses are the square root of the sum of the squares of the stresses of all cases included in the combination. This method is typically used to combine seismic directional components.

Abs

Indicates a combination of the absolute values of quantities, such as the displacements, the forces, or the stresses. Displacements are the sum of the absolute value of the displacements of all cases included in the combination. Forces are the sum of the absolute value of the forces of all cases included in the combination. Stresses are the sum of the absolute value of the stresses of all cases included in the combination. This method is typically used to combine SUS cases with OCC cases for occasional stress code check.

For flange checks that use the **Abs** load case combination method (specified in Piping Input), the software uses the following calculated (local) values:

- The axial force and torsion, which is the absolute sum of these values from all load cases included the combination.
- The bending moment, which is the sum of resultant bending moments from all load cases included in the combination.

The flange pressure, which is the maximum pressure defined in the load cases included in the combination.

NOTE For DNV, only **Abs** and **Scalar** are allowed.

Max

Indicates a combination that reports the maximum displacement, the maximum force, and the maximum stress value of the cases combined. This method retains the original sign. Displacements are the displacements having the maximum absolute values of all the load cases included in the combination. Forces are the forces having the maximum absolute values of all the load cases included in the combination. This method is typically used to report the greatest restraint loads from among a selected set of load cases.

Stresses are the stresses having the maximum absolute values of all the load cases included in the combination. When multiple expansion load cases use the **Max** combination method, the software uses the maximum code stress and corresponding allowable stress.

For flange checks that use the MAX load case combination method (specified in CAESAR II piping input), the software uses the following calculated (local) values:

- The axial force and torsion, which is the maximum magnitude of these values from all load cases included the combination.

- The bending moment, which is the maximum resultant bending moment from all load cases included in the combination.

The flange pressure, which is the maximum pressure defined in the load cases included in the combination.

Min

Indicates a combination that reports the minimum displacement, the minimum force, and the minimum stress value of the cases combined. This method retains the original sign. Displacements are the displacements having the minimum absolute values of all the load cases included in the combination. Forces are the forces having the minimum absolute values of all the load cases included in the combination. Stresses are the stresses having the minimum absolute values of all the load cases included in the combination.

SignMax

Indicates a combination that reports the maximum displacement, the maximum force, and the maximum stress value of the cases combined. The sign is considered in the comparison. Displacements are the maximum signed values of all the displacements from each case included in the combination. Forces are the maximum signed values of all the forces from each case included in the combination. Stresses are the maximum signed values of all the stresses from each case included in the combination. This method is typically used in conjunction with **SignMin** to report the envelope of restraint loads from among a selected set of load cases.

SignMin

Indicates a combination that reports the minimum displacement, the minimum force, and the minimum stress value of the cases combined. The sign is considered in the comparison. Displacements are the minimum signed values of all the displacements from each case included in the combination. Forces are the minimum signed values of all the forces from each case included in the combination. Stresses are the minimum signed values of all the stresses from each case included in the combination. This method is typically used in conjunction with **SignMax** to report the envelope of restraint loads from among a selected set of load cases.

Snubbers Active

Indicates whether snubbers are active. Select to indicate that snubbers are treated as rigid restraints for the load case. By default, Occasional (OCC) load cases activate this option while other types of load cases clear this option.

Hanger Stiffness

Specifies the hanger stiffness for the load case.

As Designed

Causes the software to consider the actual spring hanger stiffnesses. Use this option for most real (non-hanger design) load cases.

Rigid

Causes the software to model the spring hangers as rigid restraints. Use this option for restrained weight cases and hydrotest cases if the spring hangers are pinned.

Ignore

Causes the software to remove the spring hanger stiffnesses from the model. Use this option for hanger travel cases, unless you want to include the stiffness of the selected spring in the operating for hanger travel case and iterate to a solution. In that case, select **As Designed**. Also adjust the hanger load in the cold case (in the physical system) to match the reported hanger cold load.

 **NOTE** User-defined hangers are not made rigid during restrained weight cases.

Fluid Density Multiplier

Specifies the multiplier of fluid density (FDM) used in this load case. The fluid density used for an element is this multiplier times the Fluid Density specified for the element.

If you define FDM as zero, the software ignores the Fluid Density for this load case and only uses the Pipe Density and insulation densities, similar to the [WNC load type](#).

 **NOTE** This option supports Multiple Fluid Densities Using the Fluid Density Multiplier.

Elastic Modulus

Designates use of Cold (EC) or any of the nine (EH1-EH9) hot elastic moduli to determine results on a per-load-case basis.

EC

Cold elastic modulus.

EH1

Hot elastic modulus corresponding to **T1**.

EH2 - EH9

Hot elastic modulus corresponding to **T2** through **T9**.

Elbow Stiffening Pressure

Specifies the pressure used to determine the modifiers for the **SIF** and **k** factors on a per-load-case basis.

Pmax

Maximum of **P1** through **P9**.

None

No pressure stiffening for the elbow.

P1 - P9

Operating pressures 1 through 9.

Phydro

Hydrostatic pressure.

Elbow Stiffening Elastic Modulus

Specifies the elastic modulus is used to determine the modifiers for the **SIF** and **k** factors on a per-load-case basis.

EC

Cold elastic modulus.

EH1 - EH9

Hot elastic modulus corresponding to **T1** through **T9**.

SUS/OCC Case Sh

Designates the use of a hot allowable stress (S_h) to determine the results on a per-load case basis. Use this option for sustained (SUS) and occasional (OCC) load cases.

Sh_min

Minimum of **Sh1** through **Sh9**.

Sh1 - Sh9

Hot allowable stresses corresponding to **T1** through **T9**.

 **NOTE** The **SUS Case Sh** option applies only to the B31.1-2020, EN-13480-3:2017/A5:2022, and B31.3-2010 codes and later.

Friction Multiplier

Specifies the multiplier of friction factors used in this load case. The Coefficient of Friction used at a restraint is this multiplier times the Mu factor at the restraint. Set this value to zero to deactivate friction for this load case.

OCC Load Factor

Displays/overrides the Occasional Load Factor defined in the Configuration Editor. The default value changes according to the piping code and the frequency of occurrence.

ISO-14692

Occasional load factors are defined differently for different load cases (**Operating**, **Sustained**, **Occasional**, and **Hydrotest**). The default occasional load factors for these load cases are:

1.0

Sustained load cases

1.24

Operating load cases

1.33

Occasional and Hydrotest cases

The Occasional Load Factor and the System Design Factor (defined on the Allowable Stresses tab of the **Auxiliary Data** panel) are multiplied together to generate the part factor for loading (f_2) as defined in ISO-14692.

Example

A default System Design Factor of 0.67 with the above default occasional load factors results in the following default values for the part factor for loading:

Load Case Type	System Design Factor	Occasional Load Factor	Part Factor for Loading (f_2)
Sustained (SUS)	0.67	1.00	0.67
Operating (OPE)	0.67	1.24	0.83
Occasional (OCC)	0.67	1.33	0.89
Hydrotest (HYD)	0.67	1.33	0.89

Flange Analysis Temperature

Specifies the temperature used to determine the flange allowable.

None

No flange analysis.

T1 - T9

Operating temperatures **T1** through **T9**.

Tmax

Maximum of **T1** through **T9**.

TAmb

Ambient temperature.

Limit State Load Type

Specifies the load type for the DNV limit state. Define these values for non-combined and non-LMST-combined loads for use with combination load cases and the **LMST** stress type.

None

The software does not calculate any DNV limit state.

Empty

Load on empty pipe. Use **Empty** with the **WNC** weight in the load case.

Water Filled

Load on pipe during water-filled installation. Use **Water Filled** with the **WW** weight in the load case.

Test Hydro

System testing under hydrostatic pressure. Use **Test Hydro** with the **WW** weight in the load case.

Operation

Operational loading. Use **Operation** with the **W** weight in the load case.

Environment

Environmental loading. Use **Environment** with the **W** weight in the load case.

 **NOTE** These options also effect the stress calculations for bend elements through α_U .

Wind Loads Tab (Static Analysis - Load Case Editor Dialog)

Controls options for wind loads. Specify up to four different wind load profiles.

 **NOTE** For technical information, see Wind Loads.

Editing Wind Case

Specifies the wind case to edit. The first field indicates the active wind case. The second field displays the total number of defined cases.

Select Wind Code/Profile and Year

Specifies the wind code or one of the user-defined (velocity or pressure) profiles. Select the code and then select the code edition. The software updates the dialog for your code selection.

[ASCE 7 Wind Code Options](#)

[AS/NZS 1170 Wind Code Options](#)

[BS-6399-2 Wind Code Options](#)

[Brazil NBR 6123 Wind Code Options](#)

[China GB 50009 Wind Code Options](#)

[EN Wind Code Options](#)

[IBC/UBC Wind Code Options](#)

[IS 875 Wind Code Options](#)

[Mexico Wind Code Options](#)

[NBC Wind Code Options](#)

[Wind Pressure/Velocity vs. Elevation \(User-Defined\)](#)

 **NOTE** See the Wind Loads technical discussion.

Copy Wind Vector

Copies the wind data from the wind case selected for [Editing Wind Case](#) to any or all of the remaining wind cases.

NOTE Use this function for large wind pressures or with Velocity versus Elevation tables.

ASCE 7 Wind Code Options

Specifies wind load and topographic factor parameters for the ASCE 7 wind code.

NOTE For more information on using wind reduction factors with ASCE 7, see ASCE 7/IBC Wind and Seismic Load Combinations.

Basic Wind Speed

Typical units (depending on your unit system):

- ft/sec
- m/sec
- mm/sec

Specifies the three-second gust speed at 33 ft. (10 m.) above ground for Exposure C category as determined according to ASCE 7-05 Section 6.5.6.3 or ASCE 7-10/16/22 Section 26.5.1.

According to ASCE 7, the following are typical basic wind-speed values:

- California and West Coast Areas - 124.6 ft./sec. (85 mph)
- Rocky Mountains - 132.0 ft./sec (90 mph)
- Great Plains - 132.0 ft./sec (90 mph)
- Non-Coastal Eastern United States - 132.0 ft./sec (90 mph)
- Gulf Coast - 190.6 ft./sec (130 mph)
- Florida-Carolinas - 190.6 ft./sec (130 mph)
- Miami - 212.6 ft./sec (145 mph)
- New England Coastal Areas - 176.0 ft./sec (120 mph)

Wind Exposure

Specifies the wind exposure. This value is the exposure category that adequately reflects the characteristics of ground surface irregularities. The ASCE 7 code standard (found in ASCE 7-05, Section 6.5.6.3 and ASCE 7-10/16/22, Section 26.7.3) defines exposure categories as follows:

B or 2

Exposure B. Urban, suburban, and wooded areas, prevailing. For upwind distance requirements, refer to the code standard.

C or 3

Exposure C. All cases where exposures B and D do not apply.

D or 4

Exposure D. Flat coastal areas, prevailing. Urban, suburban, wooded, or open terrain areas with scattered obstructions. For upwind or downwind requirements, refer to the code standard.

 **NOTE** See ASCE 7-05 (Section 6.5. 6.3) and ASCE 7-10/16/22 (Section 26.7.3) for any exceptions on wind exposures.

Structural Damping Coefficient

Specifies the structural damping coefficient. This value is the percentage of critical damping and is used to calculate the gust factor for the wind load calculations.

Structural Classification

Specifies the classification of buildings and structures based on the type of occupancy.

ASCE 7-2005 Table 1-1, classification is as follows:

1 - Category I

Failure represents low hazard.

2 - Category II

All structures except 1, 3, and 4.

3 - Category III

Primary occupancy more than 300 people.

4 - Category IV

Essential facilities (Hospitals, and so forth)

 **NOTE** **1**, **2**, **3** and **4** are the options for structural classification categories, and are equivalent to categories **I**, **II**, **III** and **IV** defined in ASCE 7.

Importance Factor

Specifies the importance factor (I). This value is used to calculate the velocity pressure for wind load calculations. The importance factor depends on the structural classification and whether the region is prone to hurricanes, as shown below.

For ASCE 7 2010, CAESAR II sets the importance factor to **1**, as the code standard no longer addresses this factor.

For ASCE 7 2005, set the importance factor as per the code standard in Table 6-1.

Category	Non-Hurricane	Hurricane Prone
1-I	0.87	0.77
2-II	1.00	1.00
3-III	1.15	1.15
4-IV	1.15	1.15

NOTE 1, 2, 3, and 4 are the options for structural classification categories, and are equivalent to categories I, II, III and IV, defined in ASCE 7.

Structure Natural Frequency

Specifies the natural frequency used to calculate the dynamic wind gust effect factor for dynamically sensitive structures. ($f < 1$ Hz.)

If the natural frequency is not defined, CAESAR II uses a gust effect factor 0.85.

Ground (System) Elevation Above Sea Level

Specifies the ground elevation factor used to adjust for air density, determined according to Section 26.9 of ASCE 7-16 and later.

Height of Hill or Escarpment

Specifies the height of hill or escarpment value. This value is the height relative to the upwind terrain. It is used in calculations of the topographic factor of structures sited on the upper half of hills and ridges or near the edges of escarpments.

Crest Distance

Specifies the distance upwind of crest to where the difference in ground elevation is half the height of hill or escarpment.

Distance from Crest to Site

Specifies the distance upwind or downwind from the crest to the building site.

Hill Type

Specifies the hill type. This value is the hill type is defined as follows:

0

No Hill

1

2D Ridge

2

2D Escarpment

3

3D Axisymmetric Hill

AS/NZS 1170 Wind Code Options

Specifies options for the AS/NZS 1170.2 wind code.

Design Wind Speed

Typical units (depending on your unit system):

- ft/sec
- m/sec
- mm/sec

The regional wind speed V_r , described in section 3.2 of the code.

Wind Region

The wind region is determined from the geographic locations for Australia and New Zealand. The maps of these locations are in Figure 3.1 of the code.

Terrain Category

Value discussed in section 4.2.1 of the code. These categories are defined as:

Category 1

Exposed open terrain with few or no obstructions and water surfaces at serviceable wind speeds

Category 2

Water surfaces, open terrain, grassland with few, well-scattered obstructions having heights generally from 1.5 m to 10 m.

Category 3

Terrain with numerous closely spaced obstructions 3 m to 5 m high such as areas of suburban housing

Category 4

Terrain with numerous large, high (10 m to 30 m high) and closely spaced obstructions such as large city centers and well-developed industrial complexes

 **NOTE** Choose the terrain category with due regard to the permanence of the obstructions that constitute the surface roughness. Vegetation in tropical cyclonic regions cannot be relied upon to maintain surface roughness during wind events.

Lee Effect Multiplier (Mlee)

Specifies the Lee Effect Multiplier. The default value is 1.0. Paragraph 4.4.3 discusses the issue of the lee effect multiplier. In the case of New Zealand, reference is made to the New Zealand site map. For all other sites, it shall be taken as 1.0.

Hill Shape Factor (Mh)

Specifies the appropriate hill shape factor, which can be obtained from Table 4.4 of the code. Please refer to paragraph 4.4.2 which gives precise details for the derivation of the hill shape factor.

Upwind Slope	
(H/2Lu)	Mh
< 0.05	1.00
0.05	1.8
0.10	1.16
0.20	1.32
0.30	1.48
>= 0.45	1.71

Wind Direction Multiplier (Md)

Specifies the wind direction multiplier. The default value is 1.0.

The wind direction multiplier is detailed in paragraph 3.4 of the code, specifically Table 3.2. As the wind multiplier is determined from the cardinal wind directions (N, NE, E, SE, S SW, W and NW), the value for any direction is specified in the table as 1.0. We recommend this value be used for all cases.

Convert to Permissible Stress Gust Wind Speed

In the standard AS/NZS 1170.2 Supp 1:2002 Section C3, there is a discussion regarding the division of the wind speed given in the standard by the square root of 1.5. Checking the box converts the wind speed given to a permissible stress basis. Doing this lowers the wind loads on the vessel.

Surface Roughness Height (hr)

Typical units (depending on your unit system):

- in
- mm
- cm

This value is used to compute the ratio h_r/d which is then used to compute the drag force coefficient (C_d) for rounded cylindrical shapes per Table E3. For pressure vessels, this value ranges from 0.003 mm for painted metal surfaces to 15 mm for heavily rusted surfaces. Light rust has a value of 2.5 mm while galvanized steel has a value of 0.15 mm.

NOTE The ratio hr/d is taken to be unitless (mm/mm).

Site Elevation (E)

Specify the height of the site above the mean sea level, E.

Average Spacing of Shielding Buildings

Typical units (depending on your unit system):

- ft
- m
- mm

Specifies the average spacing of the shielding buildings. This is discussed in paragraph 4.3.3 of the code.

Average Breadth of Shielding Buildings (bs)

Typical units (depending on your unit system):

- ft
- m
- mm

Specifies the average breadth of the buildings that shield the piping.

Average Height of Shielding Buildings (hs)

Typical units (depending on your unit system):

- ft
- m
- mm

Specifies the average height of the buildings that shield the piping.

Number of Upwind Bldgs at 45 degs

Specifies the number of upwind buildings within a 45 degree arc. The upwind buildings are the ones shielding the piping.

BS-6399-2 Wind Code Options

Specifies options for the BS-6399-2 British wind code.

Design Wind Speed

Specifies the design value of the wind speed. These vary according to geographical location and according to company or vendor standards. Here are a few typical wind speeds in miles per hour. Typical wind speeds are shown in Figure 6 of BS-6399-2. The wind speeds are only relevant to the United Kingdom. The wind speeds vary typically from 20 m/sec to 31 m/sec (44.7 mph to 69.3 mph).

Type the lowest value reasonably allowed by the standards you are following, because the wind design pressure (and thus force) increases as the square of the speed.

Site Elevation - delta s

Specifies the site altitude above mean sea level (paragraph 2.2.2.2 of the code). Use this value plus the Base Elevation to calculate the height of each point in the vessel above mean sea level. For example, if the vessel is installed on a site that is 100 m (328 ft) above sea level, it is exposed to a higher wind pressure (P) than if installed on the beach (at mean sea level).

Upwind Building Height (Obstruction Height) - Ho

For buildings in town terrain, specifies the average height of the building upwind of the piping (as they tend to shield the piping from the wind). To be conservative, this value can be zero, so the piping takes the full force of the wind. Ho is used to modify the effective piping wind height (He) for any piping element. See paragraph 1.7.3.3 of BS-6399-2.

Upwind Building Spacing - X

For buildings in town terrain, type the average spacing of the buildings upwind of the piping (as they tend to shield the piping from the wind). If the buildings are closer together, they provide greater protection from the wind. See paragraph 1.7.3.3 of BS-6399-2.

Pipe Location

Specifies the location where the system is installed, either in the country, or in a town. The BS-6399-2 factors in Table 4 modify the wind velocity. The final wind pressure acting on any element of the piping is determined by the distance from the coast, whether located in the country or a town, and the effective height (He). This table derives Sb, which is calculated by internally.

Distance to Coastline

Specifies the distance the vessel is located from the coast in kilometers. This distance affects the corrected wind speed (V_e). The BS-6399-2 factors in Table 4 modify the wind velocity. The final wind pressure acting on any element of the vessel is determined by the distance from the coast, whether located in the country or a town, and the effective height (He). This table derives Sb, which is calculated by internally.

Factor Kb from Table 1 - Kb

Specifies the *Building-type factor Kb* taken from Table 1 of BS6399. Choose from one of five values: 8, 4, 2, 1 or 0.5. CAESAR II sets the default to 2, but any other value may be chosen. Please note the following limitations

of K_b based on the vessel height:

K_b	Maximum Vessel Total Height
8	23 m (75.4 ft)
4	75 m (246 ft)
2	240 m (787 ft)
1	300 m (984 ft)
0.5	300 m (984 ft)

Designing towers over 75 meters in height is unlikely and you would need to consider many other things.

BS 6399 Table 1. Building-type Factor K_b	
8	Welded Steel unclad frames
4	Bolted steel and reinforced concrete unclad frames
2	Portal sheds and similar light structures with few internal walls
1	Framed buildings with structural walls around lifts and stairs only (e.g. office buildings of open plan or with partitioning)
0.5	Framed buildings with structural walls around lifts and stairs with additional masonry subdivision walls (for example, apartment buildings), building of masonry construction and timber-framed housing

Annual Probability Factor - Q

Calculates the final probability factor (Q) associated with the likelihood of high velocity gusts occurring over certain periods such as 50 years. The default value is $Q = 0.02$. The code sets 0.02 as a standard value for a mean recurrence value of 50 years. Annex D of BS-6399-2 should be consulted for a fuller explanation.

Q	Explanation
0.632	NOTE 1: The annual mode, corresponding to the most likely annual maximum value. ($S_p = 0.749$)
0.227	NOTE 2: For the serviceability limit, assuming the partial factor for loads for the ultimate limit is $f = 1.4$ and for the serviceability limit is $f = 1.0$, giving $S_p = \text{Sqrt}(1 / 1.4) = 0.845$. ($S_p = 0.845$)
0.02	NOTE 3: The standard design value, corresponding to a mean recurrence interval of 50 years. ($S_p = 1.000$)
0.0083	NOTE 4: The design risk for bridges, corresponding to a mean recurrence interval of 50 years. ($S_p = 1.048$)
0.00574	NOTE 5: The annual risk corresponding to the standard partial factor for loads, corresponding to a mean recurrence interval 1754 years. This is back-calculated assuming the partial factor load for the ultimate limit is $f = 1.4$ and all risk is ascribed to the recurrence of wind. ($S_p = \text{Sqrt}(1.4)$)
0.001	NOTE 6: The design risk for nuclear installations, corresponding to a mean recurrence interval of 10,000 years. ($S_p = 1.263$)

Seasonal Factor - Ss

BS6399 in paragraph 2.2.2.4 states: "...For permanent buildings and buildings exposed for continuous periods of more than 6 months a value of 1.0 should be used for Ss..." The software uses 1.0 as the default value for this reason. Using a value of less than 1.0 is not recommended or should only be used with solid research.

Directional Factor - Sd

Taken from Table 3 of BS6399. Because a tower is symmetrical about its central axis, the default value has been taken as 1.0. It is recommended that this value not be reduced other than for exceptional circumstances. For other values, please consult Table 3. The values in that table range between 0.73 and 1.00.

Pipe Surface Type

Specifies the pipe surface condition. Select **1 Smooth**, **2 Rough**, or **3 Very Rough**.

Total Wind Height

Specifies the total height of the building or structure, which CAESAR II uses in wind force equations. For piping systems, consider using the maximum height of the piping system. Refer to the wind code standard for more information on this value.

Brazil NBR 6123 Wind Code Options

Specifies options for the Brazil NBR 6123 wind code.

Basic Wind Velocity (Vo)

Velocity from a three second gust, exceeded only once in 50 years. It is measured at 10 meters over smooth open ground and depends on the plant location. The wind may blow in any horizontal direction. This velocity is taken from Figure 1, and item 8 which shows the iso-velocities over Brazil. The referred to Figures and Tables are found in the Petrobras document BPE-500-P4-19i and the Brazilian Wind Code NBR 6123.

Topographical Factor (S1)

Accounts for the variations and profile of the land. For plain or slightly uneven ground, use a value of 1. The larger this value is, the greater the final computed wind pressure is. If the vessel is on a hill top, this value should be computed according to section 5.2 of NBR 6123.

Roughness Category (S2)

Category	Description
1	Plain ground with large dimensions (more than 5 km of extension)
2	Plain (or slightly uneven) ground with few, and separated, obstacles

3	Plain or uneven ground obstructed by obstacles (walls or separated low buildings)
4	Ground with many grouped obstacles in industrial or urban areas
5	Ground with many grouped and tall obstacles (such as developed industrial areas)

NOTE Using Category I produces a higher wind load than Category II and so forth.

Dimension Class

Class	Description
A	Greatest dimension is less than or equal to 20 meters
B	Greatest dimension is greater than 20m and less than 50 meters
C	Greatest dimension is greater than or equal to 50 meters

Statistical Factor (S3)

Accounts for security and the expected life of the equipment. For industrial plants S3 is generally taken to be 1.0.

Pipe Surface Condition

Vessel surface condition can be classified as smooth or rough. A selection of rough results in an increased value of the shape coefficient. Using a rough classification generates a higher wind load on the vessel as there is more drag. The shape coefficient is computed based on the height to diameter ratio of the vessel.

Frequency

Specifies the natural frequency used to calculate the dynamic wind gust effect factor for dynamically sensitive structures. ($f < 1$ Hz.)

If the natural frequency is not defined, CAESAR II uses a gust effect factor 0.85.

Total Wind Height

Specifies the total height of the building or structure, which CAESAR II uses in wind force equations. For piping systems, consider using the maximum height of the piping system. Refer to the wind code standard for more information on this value.

China GB 50009 Wind Code Options

For details on wind code parameters used in CAESAR II, refer to the 2002 and 2012 editions of the China GB 5009 wind code standard.

The software uses the following default values:

Frequency

If you do not define a value, the software uses 5.0 as the value for the 2012 edition.

Total Wind Height

If you do not define a value, the software uses the element's elevation as the default value for the 2012 edition.

EN Wind Code Options

Specifies options for the European EN 1991-1-4 wind code.

Ref. Wind Velocity [$V_{b,0}$]

Specifies the fundamental value of the basic wind velocity of the area where the equipment is situated. $V_{b,0}$ is used along with C Dir and C Season to compute V_b .

Terrain Category

Select the appropriate terrain category from the table below. Category 0 generates the highest wind loads while category 4 produces the lowest wind loads.

Terrain Category	Description
0	Sea or Coastal area exposed to the open sea
1	Lakes or flat and horizontal areas with negligible vegetation and without obstacles
2	Area with low vegetation such as grass and isolated obstacles (trees, buildings) with separations of at least 20 obstacle heights
3	Area with regular cover of vegetation or buildings or with isolated obstacles with separations of maximum 20 obstacle heights (such as villages, suburban terrain, permanent forest)
4	Area in which at least 15% of the surface is covered with buildings and their average height exceeds 15 m

Directionality Factor [C Dir]

Value of the directionality factor, **C Dir**, found in the National Annex. The recommended value is 1.0.

Season Factor [C Season]

Value of the season factor, **C Season**, found in the National Annex. The recommended value is 1.0.

Structural Factor [CsCd]

Structural factor used to determine the force on the vessel. This value is defined in Section of the EN 1991-1-4:2005(E) Wind load specification in Annex D. This value normally ranges between 0.90 and 1.10. The greater the structural factor value, the higher the element load.

Structure Damping Coefficient

Specifies the structural damping coefficient. This value is the percentage of critical damping and is used to calculate the gust factor for the wind load calculations.

IBC/UBC Wind Code Options

Specifies options for the UBC and IBC wind codes.

NOTE For more information on using wind reduction factors with IBC, see ASCE 7/IBC Wind and Seismic Load Combinations.

Design Wind Speed (UBC & IBC 2006)

Specifies the design value of the wind speed. This varies according to geographical location and according to company or vendor standards. Typical wind speeds in miles per hour are 85.0, 100.0, 110.0, and 120.0.

NOTE Type the lowest value reasonably allowed by the standards you are following because the wind design pressure (and thus force) increases as the square of the speed.

Basic Wind Speed (IBC 2018)

Specifies the basic full wind speed, V , according to paragraph 1609.3.1, Equation 16-33.

NOTE See ASCE 7/IBC Wind and Seismic Load Combinations for wind load reductions, which should only be applied to the load cases.

Exposure Constant (UBC & IBC 2006) / Wind Exposure (IBC 2018)

Specifies the exposure factor, as defined in UBC-91 Section 2312 or the IBC Exposure Constant:

Exposure B

Terrain with building, forest, or surface irregularities 20 feet or more in height covering at least 20 percent or the area extending one mile or more from the site.

Exposure C

Terrain which is flat and generally open, extending one-half mile or more from the site in any full quadrant.

Exposure D

The most severe exposure with basic wind speeds of 80 mph or more. Terrain which is flat and unobstructed facing large bodies of water over one mile or more in width relative to any quadrant of the building site. This exposure extends inland from the shoreline 1/4 mile or 0 times the building (vessel) height, whichever is greater.

NOTES

- The options **Not Used** and **Exposure A** are available for Exposure Constant under IBC 2006, but they are not supported according to the codebook. Users should avoid selecting either option.

- Most petrochemical sites use a value of 3, exposure C. This value is used to set the Gust Factor Coefficient (Ce) found in Table 23-G.

Importance Factor (IBC)

Specifies the occupancy importance factor, IE, as defined in Section 1616.2 and shown in Table 1604.5. The calculated spectrum accelerations will be multiplied by this value to generate the shock spectra. Values range from 1.0 to 1.25 based on the function of the structure.

Importance Factor (UBC)

Specifies the importance factor. The software uses this value directly without modification. This value is taken from Table 23-L of the UBC standard. Followings are the context of Table 23-L:

Category	Value
I - Essential facilities	1.15
II - Hazardous facilities	1.15
III - Special occupancy structures	1.00
IV - Standard occupancy structures	1.00

Height of Hill (UBC & IBC 2006) / Height of Hill or Escarpment (IBC 2018)

Specifies the height of hill or escarpment value. This value is the height relative to the upwind terrain. It is used in calculations of the topographic factor of structures sited on the upper half of hills and ridges or near the edges of escarpments.

Distance to Site (IBC 2006) / Distance from Crest to Site (IBC 2018)

Specifies the distance upwind or downwind from the crest to the building site.

Crest Distance (IBC)

Specifies the distance upwind of crest to where the difference in ground elevation is half the height of hill or escarpment.

Ground (System) Elevation Above Sea Level (IBC 2018)

Specifies the ground elevation factor used to adjust for air density, determined according to Section 26.9 of ASCE 7-16 and later.

Hill Type

Specifies the hill type. This value is the hill type is defined as follows:

0

No Hill

1

2D Ridge

2

2D Escarpment

3

3D Axisymmetric Hill

Structure Damping Coeff. (IBC) / Structure Damping Coeff. (Beta) (UBC)

Specifies the structural damping coefficient. Type the value of structural damping coefficient (percentage of critical damping) beta. The default value is 0.01.

This value is used to compute the dynamic gust effect factor G as outlined in the commentary section 6.6 page 158 of ASCE 95 or section 6.5.8 pages 29-30 of ASCE 98. For other design codes, then the software uses the damping coefficient for that wind design code.

If your design specification does not call out for a specific value of beta, then use the default value of 0.01. Other values of beta can be specified for the filled case and the empty case. Again, if the specifications do not supply these values for empty and filled, then leave these cells blank.

Natural Frequency (IBC 2006) / Structure Natural Frequency (IBC 2018)

Specifies the natural frequency used to calculate the dynamic wind gust effect factor for dynamically sensitive structures. ($f < 1$ Hz.)

If the natural frequency is not defined, CAESAR II uses a gust effect factor 0.85.

IS 875 Wind Code Options

Specifies options for the Indian Standard, IS-875 (Part 3), wind code. IS-875 is the standard in India for design loads (other than earthquake) for buildings and structures.

Basic Wind Speed

Basic wind speed as applicable to 10 m height above mean ground level for different zones in the country can be directly calculated if the proper value is defined by the **Wind Zone Number**. Alternatively, wind speed can be defined here. Basic wind speed should be based on peak gust velocity averaged over a short time interval of about 3 seconds and correspond to mean heights above ground level in an open terrain. This option is optional.

Wind Zone Number

Figure 1 of IS-875 shows different Wind Zones of the country. Various zone numbers and corresponding Basic Wind Speed values are:

Zone 1:	33 m/sec 73.82 miles/hour
Zone 2:	39 m/sec 87.25 miles/hour
Zone 3:	44 m/sec 98.43 miles/hour

Zone 4:	47 m/sec 105.15 miles/hour
Zone 5:	50 m/sec 111.86 miles/hour
Zone 6:	55 m/sec 123.04 miles/hour

The value typed here must be between 1 and 6. The zone the vessel is in is determined from a map of India showing the various wind zones.

Optionally, define the basic wind speed directly in the **Basic Wind Speed** box. If the wind speed is specified, it overrides the value of wind speed and zone based on the table above.

Risk Factor (K1)

Assuming the mean probable design life as 100 years, the corresponding risk coefficient values for various wind zones are:

Zone	K1
Zone 1	1.05
Zone 2	1.06
Zone 3	1.07
Zone 4	1.07
Zone 5	1.08
Zone 6	1.08

Terrain Category

Terrain in which specific equipment is assessed as one of the following categories:

Category 1

Exposed open terrain with few or no obstructions and in which the average height of any object surrounding the equipment is less than 1.5 m. This category includes open sea - coasts and flat treeless plains.

Category 2

Open terrain with well scattered obstructions having height generally between 1.5 to 10 m. This includes airfields, open parklands and undeveloped sparsely built up outskirts of towns and suburbs. This category is commonly used for design purpose.

Category 3

Terrain with numerous closely spaced obstructions having the size of buildings and structures up to 10 m in height. This includes well wooded areas, towns, and industrial areas full or partially developed.

Category 4

Terrain with numerous tall, closely spaced obstructions. This includes large city centers, generally with obstructions above 25 m, and well developed industrial complexes.

Structure Damping Coefficient (Beta)

Specifies the structural damping coefficient. This value is the percentage of critical damping and is used to calculate the gust factor for the wind load calculations.

Equipment Class

Equipment and structures are classified into following classes depending upon their size.

Class A

Equipment and components having a maximum dimension (greatest horizontal or vertical dimension) less than 20 m.

Class B

Equipment and components having a maximum dimension (greatest horizontal or vertical dimension) between 20 and 50 m.

Class C

Equipment and components having a maximum dimension (greatest horizontal or vertical dimension) greater than 50 m.

Topography Factor

The topography factor ranges between 1.0 and 1.36. This factor takes care of local topographic features such as hills, valleys, cliffs, ridges and so on, which can significantly affect wind speed in their vicinity. The effect of topography is to accelerate wind near summits of hills or crests of cliffs and decelerate the wind in valleys or near the foot of cliffs. Effect of topography is significant if upwind slope is greater than about 3 degrees. Below 3 degrees, the value of K3 can be taken as 1.0. For slopes above 3 degrees, the value of K3 ranges between 1.0 and 1.36.

Use Gust Response Factor

Specifies that the software calculates the gust response factor per IS-875 and uses the factor in the appropriate equations. Experience has shown that these gust response factors are very conservative. Select this option only if the design specifications and the customer or owner explicitly require you to do so.

Mexico Wind Code Options

Specifies the options available for the Mexico wind code.

Design Wind Speed

According to Paragraph 4.6.2, Isotach Maps. Regional Velocity (VR), the velocity of the wind, VR, is the maximum mean velocity likely to occur within a certain recurrence period in a determined zone or region of the country.

The Isotach maps that are included in this clause with the different periods of return, such velocities refer to homogenous conditions that correspond to a height of 10 meters over the surface of the floor in the flat

terrain (Category 2 per Table I.1). It does not consider the local terrain roughness characteristics or the specific topography of the site. Therefore, the velocity is associated with three-second wind gusts and considers the possibility that there might be hurricane winds present in the coastal zones.

The software determines the regional velocity, V_R , by considering the geographic location of the site of the building's uproot and its destination.

In Figures I.1 through I.4, the Isotach regional maps are shown, corresponding to the periods of recurrence for 200, 50, and 10 years.

The importance of the structures (Paragraph 4.3) dictates the periods of recurrence which should be considered for the wind design. From this, the groups A, B and C associate themselves with the periods of return of 200, 50 and 10 years, respectively. The uproot site is in the map with the recurrence period which corresponds to the group to which the building belongs to, in order to obtain the regional velocity. In the Tomo III from Ayudas de Dise O a table is shown with the main cities in the country and their corresponding regional velocities for the different periods of return.

Structural Classification

Specifies the structural classification.

Class	Description
A	Every remote structural element exposed directly to the wind action. Horizontal or vertical structures that measure less than 20 meters of length.
B	Horizontal or vertical structures that measure between 20 and 50 meters of length.
C	Horizontal or vertical structures that measure more than 50 meters of length.

Terrain Category

Defined in Table I.1, based on the type of soil and roughness.

Category	Description
1	Open terrain, practically smooth, without obstructions.
2	Flat or undulating soil, with few obstructions.
3	Terrain covered by many obstructions narrowly spaced.
4	Terrain with many big, tall, narrowly spaced obstructions.

Topographic Factor (Ft) (Paragraph 4.5.4)

This factor considers the local topographic effect from the place in which the structure uproots. For example, if the building is found on the hillsides, on top of hills, or on mountains at important heights with respect to the general level of the terrain of its outskirts, it is probable that wind generates accelerations and the regional velocity should be increased.

Damping Factor (Zeta)

This value is typically 0.01.

Strouhal Number (St)

The Strouhal number is unitless; 0.2 for circular sections and 0.14 for rectangular sections.

Barometric Height (Omega)

Typical units (depending on your unit system):

- ft
- m
- mm

Indicates the barometric pressure in mm Hg, as shown in the following table:

Height	Barometric Pressure (mm Hg)
0	760
500	720
1000	675
1500	635
2000	600
2500	565
3000	530
3500	495

Ambient Temperature

Indicates the ambient temperature in degrees Celsius.

Frequency

Specifies the fundamental frequency time interval, in Hz.

Total Wind Height

Specifies the total height of the building or structure, which CAESAR II uses in wind force equations. For piping systems, consider using the maximum height of the piping system. Refer to the wind code standard for more information on this value.

NBC Wind Code Options

Specifies the options available for the NBC (National Building Code of Canada) wind code.

Reference Wind Pressure or Design Wind Speed

Specifies the NBC reference wind pressure. The reference velocity pressure q is the appropriate value determined in conformance with Subsection 1.1.3 (based on probability) or Table C-1.

Specifies the design value of the wind speed. This varies according to geographical location and according to company or vendor standards. Typical wind speeds in miles per hour are 85.0, 100.0, 110.0, and 120.0.

NOTE Type the lowest value reasonably allowed by the standards you are following because the wind design pressure (and thus force) increases as the square of the speed.

Importance Factor (I_w)

Specifies the NBC importance factor. This value is the importance factor for wind taken from the table below. This is table 4.1.7.1 on page 4-17 of Division B of NBC 2005.

Importance Category	Importance Factor, I_w	
	ULS	SLS
Low	0.8	0.75
Normal	1.0	0.75
High	1.15	0.75
Post Disaster	1.25	0.75

Exposure Constant

Based on the profile of mean wind speed, the **Exposure Constant** varies considerably with the general roughness of the terrain over which the wind has been blowing before it reaches the building. These exposures are applicable only under the dynamic (detailed) method, whereas the terms open and rough were used for the static (simplified) method.

Exposure A

Indicates open-level terrain (open or standard exposure) with only scattered buildings, trees, or other obstructions, open water or shorelines. Reference wind speeds are based on Exposure A.

Exposure B

Indicates rough terrain (rough exposure), such as large cities with several and dense high-rise buildings, which persists in the upwind direction for at least 20 times the building height. Prior to NBC 2010, referred to winds that persist in the upward direction for 1.0 km or 10 times the building height, whichever is larger.

Exposure C

(NBC 2005 only) - Rough terrain, such as large cities with several and dense high-rise buildings, which persists in the upwind direction for at least 1.0 km or 10 times the building height, whichever is larger.

Roughness Factor

Specifies the Roughness Factor:

- 1 - Round, moderately smooth
- 2 - Round, rough ($D'/D = 0.02$)
- 3 - Round, very rough ($D'/D = 0.08$)

Height of the Windward Face

Specifies the height of a piping section that is exposed to wind blow.

Structural Damping Coefficient

Specifies the structural damping coefficient. This value is the percentage of critical damping and is used to calculate the gust factor for the wind load calculations.

Natural Frequency

Specifies the natural frequency used to calculate the dynamic wind gust effect factor for dynamically sensitive structures. ($f < 1$ Hz.)

If the natural frequency is not defined, CAESAR II uses a gust effect factor 0.85.

Height of Hill or Escarpment

Specifies the height of hill or escarpment value. This value is the height relative to the upwind terrain. It is used in calculations of the topographic factor of structures sited on the upper half of hills and ridges or near the edges of escarpments.

Crest Distance

Specifies the distance upwind of crest to where the difference in ground elevation is half the height of hill or escarpment.

Distance from Crest to Site

Specifies the distance upwind or downwind from the crest to the building site.

Hill Type

Specifies the hill type. This value is the hill type is defined as follows:

0

No Hill

1

2D Ridge

2

2D Escarpment

3

3D Axisymmetric Hill

Wind Pressure/Velocity vs. Elevation (User-Defined)

Allows you to indicate your own wind profiles, based on one of two options:

- Wind Pressure vs. Elevation
- Wind Velocity vs. Elevation

When you select a user-defined wind code (**Wind Pressure vs. Elevation** or **Wind Velocity vs. Elevation**), enter the corresponding pressure or velocity values. Enter a single entry in the table if a uniform pressure or velocity is to act over the entire piping system. Otherwise, type the pressure or velocity profile for the applicable wind loading.

 **NOTE** The software acquires the pressure/velocity and elevation values from the piping input. They do not necessarily match the current setting in the configuration file.

Wind Direction Specification

Specifies the direction vector (cosine) which defines the direction of the wind. The magnitude of the vector is not significant. For example:

- Wind in X direction, vector is 1, 0, 0
- Wind in Z direction, vector is 0, 0, 1
- Wind at 45 degrees, vector is .707, 0, .707

You only need to specify the method and the wind direction if you are using a pressure or velocity versus elevation table. After selecting **User Wind Profile**, a dialog prompts you for the corresponding pressure or velocity table. You only need to make a single entry in the table if a uniform pressure or velocity is to act over the entire piping system. Otherwise, type the pressure or velocity profile for the applicable wind loading.

Wave Loads Tab (Static Analysis - Load Case Editor Dialog)

Controls options for wave loads. Specify up to four different wave load profiles.

 **NOTE** For technical information, see Hydrodynamic (Wave and Current) Loading.

Editing Wave Case

Specifies the wave case to edit. The first box indicates the active wave case. The second box displays the total number of defined cases.

Copy Wave Vector

Copies the wave data from the wave case selected for [Editing Wave Case](#) to any or all of the remaining wave cases.

Current Profile Type

Specifies the means of modeling the current speed against the depth profile.

Power Law

Current speed decays with depth to the 1/7 power.

Linear Table

Define the depth versus. speed table.

Linear

Current speed decays linearly with depth becoming zero at the sea bottom.

Surface Velocity

Specifies the current speed at the free surface elevation, excluding the wave. This value is superseded by the entries in a depth versus speed table.

Typical units (depending on your unit system):

- ft/sec
- m/sec
- mm/sec

Direction Cosines

Specifies the X- and Z- cosines defining the direction of the current. The current direction may differ from the direction of any accompanying wave.

Wave Theory

Specifies the wave theory by which to model any wave effects. The available theories are:

Stream Function

Dean's stream function theory.

Stream Function, Modified

Dean's stream function theory modified to include a shear current. This shear current is assumed to vary linearly from the surface speed to the bottom speed. Therefore, this option only works with the **Linear** current profile.

STOKE'S 5th

Stoke's 5th order wave theory.

STOKE'S 5th, Modified

Stoke's 5th order wave theory modified to address particle data above the mean sea level.

AIRY

Basic linear wave theory.

AIRY, Modified

Basic linear wave theory modified to address particle data above the mean sea level.

Stream Function Order

Specifies the order of the stream function when using the stream function wave theory. Typical values are from 5-21.

Water Depth

Specifies the water depth at this location.

Typical units (depending on your unit system):

- ft
- m
- mm

Wave Height

Specifies the wave height (the crest to trough distance).

Typical units (depending on your unit system):

- ft
- m
- mm

Wave Period

Specifies the wave period. That is, the time it takes for successive crests to pass a fixed reference point.

Typical units: sec.

Wave Kinematics Factor

Specifies the wave kinematics factor. According to Section 2.3.1b of API RP 2A-WSD "Recommended Practice for Planning, Designing and Constructing Fixed Offshore Platforms -- Working Stress Design", the horizontal wave velocities calculated by the Stream Function or Stokes 5th wave theories may be multiplied by this factor in order to fit them to the wave spreading and other irregularities associated with real world wave characteristics.

Typical ranges for this factor are 0.85 to 0.95 for tropical storms and 0.95 to 1.0 for extra-tropical storms. For recommendations for Gulf of Mexico and other U.S. waters, refer to Sections 2.3.4d.1 and 2.3.4f.1 of API RP 2A-WSD.

Wave Direction Cosines

Specifies the X- and Z- cosines defining the direction of the wave. The wave direction may differ from the direction of any accompanying current.

Phase Data

Specifies a wave phase.

Same Phase at All Elements

All elements of the model are simultaneously loaded with the same wave phase, typically the phase of maximum loading.

Phase Specified at Origin

Each element experiences a different loading phase, based upon its location relative to the model origin.

Phase Angle

Specifies the wave phase angle to use to calculate the wave loadings at either:

- every element
- the model origin

Free Surface Elevation

Specifies the elevation of mean sea level, in terms of model elevation. This submerges the elements of the model to the appropriate level.

Typical units (depending on your unit system):

- ft
- m
- mm

 **NOTE** Only define **Free Surface Elevation** for wave case 1. The software uses the same value for wave cases 2, 3, and 4.

Kinematic Viscosity

Specifies the kinematic viscosity of the fluid.

Typical values for seawater are shown below:

Temp (F)	$\nu(\text{in}^2/\text{sec})$	Temp (C)	$\nu(\text{mm}^2/\text{sec})$
60	1.81e-3	15.556	1.171
50	2.10e-3	10.000	1.356
40	2.23e-3	4.444	1.440

30	2.88e-3	-1.111	1.858
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Density

Specifies the density of the sea water. A typical value for salt water is 0.037 lb/in³ or 0.00103 kg/cm³.

Typical units (depending on your unit system):

- lb/in³
- kg/cm³
- kg/dm³
- kg/m³

Current Table Depth

Displays the depth values. When using Linear Table Current Model, type up to 10 depths. A value of 0.0 indicates the surface. Positive numbers indicate distance downward from the surface.

Typical units (depending on your unit system):

- ft
- m
- mm

Current Table Velocity

Displays the velocity values. When using Linear Table Current Model, type up to the current speeds corresponding to the specified depths. Current speed typed in this table overrides the **Surface Velocity** value.

Typical units (depending on your unit system):

- ft/sec
- m/sec
- mm/sec

Chart

Displays the data for the current wave load in a graph.

Chart Selection

Specifies the graph type. **Applicable Wave Theory** is the default.

Information

Displays wave data used for the graph.

View Data Table

Displays the wave data in a table.

 **NOTE** To return to the wave load data, select **Edit**.

Check for errors

1. From Piping Input, select [Start Run](#) . When error checking completes, the [Errors and Warnings Tab](#) displays.
2. For each error message (in red), select the row. The graphic view selects and zooms in to the affected element.
3. Select the [Classic Piping Input Tab](#) and make corrections to the element to resolve the error.
4. Switch between the [Errors and Warnings Tab](#) and the [Classic Piping Input Tab](#) until all errors are resolved.
5. Select [Start Run](#)  again. If errors remain, repeat the previous three steps.

Build load cases

Define up to 999 [load cases](#).

Add recommended load cases

1. [Check for errors](#).
2. From the [Classic Piping Input Tab](#), select Edit Static Load Cases  to display the [Static Analysis - Load Case Editor Dialog](#). Loads based on the parameters specified in Piping Input display in the [left pane](#).
3. Select [Recommend Load Cases](#)  to display the [Basic and Combination Load Cases](#) and [stress types](#) that satisfy the model parameters and the basic requirements of the selected piping code.
4. To delete, select the load case row, and then select **Delete a load case** .
5. To modify, change the values of any field, including the [Definition](#).

Add a new basic load case

1. Select a row and select **Add a load case** . The software adds a new row after the selected row.
2. Select one or more loads in the [Loads Defined in Input](#) pane and drag the loads to the [Definition](#) field. The software adds the loads.

-OR-

Type the load names with an operator (usually +) between them in the [Definition](#) field.
3. Type a [Name](#) and select a [Stress Type](#).
4. Define other fields as needed.

5. Select **Save**.

Add a new combination load case

Combination load cases must follow the basic load cases.

1. Select the last row and select **Add a load case** . The software adds a new row after the last row.
2. Select one or more [Load Cases](#) and drag the load cases to the [Definition](#) field. Select the [Combination Method](#).
3. If needed, select one or more loads in the [Loads Defined in Input](#) pane and drag the loads to the [Definition](#) field
4. Type a [Name](#) and select a [Stress Type](#).
5. Define other fields as needed.
6. Select **Save**.

Edit multiple load cases

Use the **Group Edit** view on the [Load Cases Tab \(Static Analysis - Load Case Editor Dialog\)](#) to update the values of multiple load cases at one time.

1. After running select [Error Check](#)  in Piping Input, select Edit Static Load Cases .

-OR-

When opening an existing error-checked model, select Static Analysis  in the Main Window.

2. Select [Load Cases Tab \(Static Analysis - Load Case Editor Dialog\)](#) in the upper-right corner of the [Static Analysis - Load Case Editor Dialog](#).
3. Select **Group Edit**.
4. SHIFT+click to select a group of load cases or CTRL+click to select discontinuous load cases.

Fields common to the selected load cases are available to edit in the right column.

5. Edit fields as needed.
6. Select **List** to return to the list view.
7. Select **Save**.

Define wind data

1. In Piping Input, double-click Wind / Wave to open the **Wind Wave** tab of the **Auxiliary Data** panel.
2. Select Wind and specify the Wind Shape Factor on the **Wind Wave** tab.
3. [Check for errors](#).

4. Select Edit Static Load Cases  to display the [Static Analysis - Load Case Editor Dialog](#). Loads based on the parameters specified in Piping Input display in the [left pane](#), including wind loads **WIN1**, **WIN2**, **WIN3**, and **WIN4**.
5. Select the [Wind Loads Tab \(Static Analysis - Load Case Editor Dialog\)](#) to first define **WIN1**, shown as **1** for [Editing Wind Case](#).
6. Under [Select Wind Code/Profile and Year](#), select a wind code and the code edition to display the code parameters.
7. Define the wind code parameters.
 - [ASCE 7 Wind Code Options](#)
 - [AS/NZS 1170 Wind Code Options](#)
 - [BS-6399-2 Wind Code Options](#)
 - [Brazil NBR 6123 Wind Code Options](#)
 - [China GB 50009 Wind Code Options](#)
 - [EN Wind Code Options](#)
 - [IBC/UBC Wind Code Options](#)
 - [IS 875 Wind Code Options](#)
 - [Mexico Wind Code Options](#)
 - [NBC Wind Code Options](#)
 - [Wind Pressure/Velocity vs. Elevation \(User-Defined\)](#)
8. Define the [wind load direction vectors](#).
9. To specify additional wind loads, select the load case number for [Editing Wind Case](#) and repeat the previous steps. Define up to four wind loads.
10. To copy values from one wind load to another, select [Copy Wind Vector](#).
11. Select **Save**.

Define wave data

1. In Piping Input, double-click Wind / Wave to open the **Wind Wave** tab of the **Auxiliary Data** panel.
2. Select Wave on the **Wind Wave** tab.
3. [Check for errors](#).
4. Select Edit Static Load Cases  to display the [Static Analysis - Load Case Editor Dialog](#). Loads based on the parameters specified in Piping Input display in the [left pane](#), including wave loads **WAV1**, **WAV2**, **WAV3**, and **WAV4**.

5. Select the [Wave Loads Tab \(Static Analysis - Load Case Editor Dialog\)](#) to first define **WAV1**, shown as **1** for [Editing Wave Case](#).
6. Define the wave parameters.
7. Define the [wind load direction vectors](#).
8. To specify additional wave loads, select the load case number for [Editing Wave Case](#) and repeat the previous steps. Define up to four wave loads.
9. To copy values from one wave load to another, select [Copy Wave Vector](#).
10. Select **Save**.

Define static seismic data

1. In Piping Input, double-click Uniform Loads to open the **Uniform Loads** tab of the **Auxiliary Data** panel.
2. For each uniform load vector, select **G's** and define **UX**, **UY**, and **UZ** on the **Uniform Loads** tab. Specify up to nine vectors.
3. [Check for errors](#).
4. Select Edit Static Load Cases  to display the [Static Analysis - Load Case Editor Dialog](#). Loads based on the parameters specified in Piping Input display in the [left pane](#), including uniform loads **U1**, **U2**, and so on for each uniform load.
5. [Build load cases](#) for the uniform loads. See [Static Seismic Load Cases](#).
6. Select **Save**.

Run the static analysis

Run a static analysis and view results in the New Analysis Reviewer

1. Check that Preferred Output Reviewer is set to the default value of **CAESAR II** in the Configuration Editor.
2. From the Main Window:

If [Error Check / Start Run](#)  has previously been run in Piping Input, select New Analysis Reviewer .
3. From Piping Input:

Select [Error Check / Start Run](#) , correct any errors, then select View Static Results 

-OR-

Select [Batch Run](#)  to run both the error check and static analysis.
4. From the [Static Analysis - Load Case Editor Dialog](#), select **Analyze**.

Run a static analysis and view results in the Static Output Processor

1. Set Preferred Output Reviewer to **Classic Output Processor** in the Configuration Editor.
2. From the Main Window:

If [Error Check / Start Run](#)  has previously been run in Piping Input, select Static .

3. From Piping Input:

Select [Error Check / Start Run](#) , correct any errors, then select View Static Results .

-OR-

Select [Batch Run](#)  to run both the error check and static analysis.

 **TIP** When Preferred Output Reviewer is set to the default **New Analysis Reviewer**, View Static Results  can be manually run at any time to open the Static Output Processor.

4. From the [Static Analysis - Load Case Editor Dialog](#), select **Analyze**.

Individual Solution Iteration

For each iteration and each load case, the software performs a static analysis finite element solution by combining element stiffnesses to form a global system stiffness matrix. Each [basic load case](#) defines a set of loads for the ends of all the elements. These elemental load sets combine into system load vectors. Using the relationship of force equals stiffness times displacement ($F=KX$), the software calculates unknown system deflections and rotations. The known deflections may change during the analysis as hanger sizing, nonlinear supports, and friction all affect both the stiffness matrix and load vectors.

CAESAR II uses the root solution from this equation, the system-wide deflections and rotations, and the element stiffnesses to determine the global X, Y, Z forces and moments at the end of each element. The software translates these forces and moments into a local coordinate system for the element, and then calculates the code-defined stresses. Forces and moments on anchors, restraints, and fixed displacement points are summed to balance all global forces and moments entering the node. Algebraic combinations of the basic load cases pick up this process where appropriate at the displacement, force and moment, or stress level.

 **NOTE** Only one iteration per load case is required for a linear system..

Solution Iterations for Nonlinear Systems

A system with non-linear effects cannot be solved explicitly by the structural algorithm. The software solves the effects through iteration until an acceptable solution is found. CAESAR II simulates the effects of non-linear restraints through an iterative process aimed at convergence of the non-linear restraints in legitimate states, such as:

- With the pipe lifted off at a one-way support and with the support function removed from the analysis.
- With the pipe sliding along a frictional restraint and with an appropriate force applied along that line of action in the analysis.

After the software completes the setup for the solution, it repeats the calculation of the displacements and rotations for each of the basic load cases. During this step, the software displays the **Incore Solver** dialog.

NOTE The dialog only displays briefly on small models.

The dialog monitors the static analysis. The upper-left section lists the number of equations to be solved and the bandwidth of the matrix that holds these equations. Multiplying the number of equations by the bandwidth gives a relative indication of the job size. The iteration count and the current case number show how much work the software has completed. Load cases with nonlinear restraints can require several solutions or iterations before the software confirms the changing assumptions about the restraint configuration. The bar graphs indicate where the software is in an individual solution.

During processing, the right side of the dialog provides information regarding the status of nonlinear restraints and hangers in the job. For example, the software displays messages noting the number of restraints that have yet to converge or any hangers that appear to be taking no load. The **Status** frame lists the element for which the forces and stresses are being calculated. Step through nonlinear restraint status on an individual basis by pressing the **F2** through **F4** keys.

Non-Convergence of the Solution

Problems with a large number of non-linear restraints may take time to converge, or may not converge at all. In this case, carefully review the system (and the status and options of the **Incore Solver**) to determine which of these effects may be excluded, or which convergence tolerances may be changed.

Post Processing

After the analysis of the system deflections and rotations, CAESAR II post-processes the results to calculate the local forces, moments, and stresses for the [basic load cases](#) and all results for the algebraic [combination load cases](#) (for example L1-L2). The software stores the total system results in multiple files.

Control analysis results

Set [Output Status](#) and [Output Type](#) on the [Load Cases Tab](#) to control the display of load case results.



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